



Quantum Protocols for Topological Codes with ZX Calculus

Salvador Márquez Grima



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Salvador Márquez Grima

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Tutorizado por:

Prof. Jara Juana Bermejo Vega

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Salvador Márquez Grima

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Abstract

The extreme fragility of quantum information against decoherence demands the development of fault-tolerant quantum computing architectures. Among the most promising proposals, we find topological codes, such as the planar surface code, whose physical constraints of two-dimensional nearest-neighbour connectivity as well as the transversality condition impose the use of lattice surgery techniques for the execution of multi-qubit logical operations. Given that the standard circuit model is structurally deficient in capturing the non-unitary dynamics of lattice surgery, we adopt ZX calculus as a unifying framework. To justify said framework, quantum protocols such as quantum teleportation, the CHSH non-local game, and the Mermin-Peres Magic Square game, are computationally implemented and verified using the PyZX library. The validation of those protocols, as well as their optimization, is obtained from the iterative application of simplification routines based on the theory of ZX calculus.

Resumen

La extrema fragilidad de la información cuántica frente a la decoherencia exige el desarrollo de arquitecturas de computación cuántica tolerantes a fallos. Entre las propuestas más prometedoras destacan los códigos topológicos, como el código de superficie planar, cuyas restricciones físicas de conectividad bidimensional a primeros vecinos (2DNN), junto con la condición de transversalidad, imponen el uso de técnicas de cirugía de red (lattice surgery) para la ejecución de operaciones lógicas multi-qubit. Dado que el modelo de circuitos estándar resulta estructuralmente deficiente para capturar la dinámica no unitaria inherente a la cirugía de red, se adopta el cálculo ZX como un marco unificador. Para justificar dicho marco, se han implementado y verificado computacionalmente protocolos cuánticos —tales como la teleportación cuántica, el juego no local CHSH y el juego del Cuadrado Mágico de Mermin-Peres— empleando la librería PyZX. La validación de estos protocolos, así como su optimización, se obtiene a partir de la aplicación iterativa de rutinas de simplificación fundamentadas en la teoría del cálculo ZX.

1 | Introduction

The theory of quantum error correction has not always been a major field inside quantum computation [23]. The theoretical landscape shifted when Peter Shor introduced his factoring algorithm, where the profound algorithmic breakthrough faced the skepticism of contemporary physicists about the physical viability of quantum computers. This skepticism was entirely justified, as quantum states are notoriously fragile, and the prospect of building a large-scale quantum system with sufficiently low error rates was regarded as impossible.

Classical error correction protocols inherently rely on redundancy, usually achieved by copying information into ancillary bits and applying methods like the majority voting procedures [19]. The principles of quantum mechanics strictly forbid this approach, as the no-cloning theorem states that an unknown quantum state cannot be duplicated. Furthermore, classical error correction requires accessing the state of the system, while in a quantum system the measurement would imply the collapse of the wavefunction and hence the loss of quantum information. Finally, while classical errors are completely discrete, quantum errors are continuous transformations, which certainly implies a higher difficulty to recover the state prior to the error.

The paradigm drastically transformed in the mid-1990s, as Peter Shor [27] and Andrew Steane [1] discovered the first quantum error-correcting codes, based on the application of specific multi-qubit operators that always yield binary results. This allowed the continuum of possible errors to collapse into a discrete set of correctable faults without accessing the state of the system.

The field of quantum error correction has developed towards the paradigm of fault-tolerant quantum computing, i.e., the design of large-scale quantum processors capable of performing quantum error correction procedures to achieve arbitrarily low error rates. The foundational principles of building reliable logical systems from inherently unreliable physical components with the study of structure invariants were established in the mid-20th century. Notably, John von Neumann's work on probabilistic logics and the synthesis of reliable macroscopic automata from unreliable components [34], as well as Moore and Shannon's construction of reliable circuits using less reliable relays [39].

As classical computational demands grew, the necessity to manage increasing physical complexity produced the Very Large-Scale Integration (VLSI) revolution, pioneered by methodologies such as the Mead-Conway design rules [33], allowing engineers to abstract the physical components into standardised logical cells. The separation of the logical kernel from the physi-

cal layout enables a massive reorganization of computational workflows. Today, fault-tolerant quantum computing is undergoing an analogous historical transition, as the research shifts from the strictly empirical approach to a tightly integrated technology-design co-optimization methodology.

The goal of the present work is completely framed inside this transition. Specifically, we want to implement ZX calculus, a tool for diagrammatic reasoning, as a compiler for quantum protocols in fault-tolerant architectures.

The thesis will be structured as follows. First, in chapter 2, we will review some basic concepts about quantum computation, building the foundational basis for the discussion later. In chapter 3, we will give a broad overview of the theory of quantum error correction, as well as the approach for the encoding of logical qubits represented in the stabilizer codes. Chapter 4 will show how stabilizer formalism and the homological approach of algebraic topology allow for the construction of a specific family of quantum error-correcting codes, of enormous relevance in current fault-tolerant quantum computing, as they respect strict two-dimensional nearest-neighbour connectivity between physical qubits. Later, in chapter 5 we will present how we can implement multi-qubit gates acting on the logical qubits defined in the previous chapter, and how this set of operations is naturally captured by ZX calculus in chapter 6. Finally, chapter 7 presents the practical problem, the technical details of the implementation using the python library PyZX, and chapter 8 show the results of said practical problem.

2 | Introduction to Quantum Computation

While in classical computation bits, the basic unit of classical information, take two possible values (0 or 1), quantum computation relies on a basic unit of information, called the qubit, that exhibits quantum behaviour [35]. In contrast to the bit, the state of a qubit can be described as a superposition of the two states of the computational basis, conventionally denoted $|0\rangle$ and $|1\rangle$, so the state of a qubit is given by:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1 \quad (2.1)$$

Therefore, the state of a qubit is described by a vector in a two-dimensional complex vector space. While we can access the information stored in a classical bit at any time, quantum mechanics impose a restriction for the case of the qubit. Foremost, when we make a measurement (in the computational basis) we do not obtain $|\psi\rangle$, but rather one of the eigenvalues of $|0\rangle$ or $|1\rangle$, with probabilities $|\alpha|^2$ and $|\beta|^2$ respectively. Furthermore, the result of the measurement also determines the collapse of the physical state of the qubit into one of the vectors of the computational basis, meaning that if we measure the qubit and obtain the eigenvalue corresponding to $|0\rangle$, the complete state of the qubit is now $|\psi\rangle = |0\rangle$. The probabilistic characterisation of measurement introduces a non-deterministic element in the dynamics of the qubit, and therefore in quantum computation in general.

Because $|\alpha|^2 + |\beta|^2 = 1$, one may rewrite equation 2.1 as:

$$\begin{aligned} |\psi\rangle &= e^{i\gamma} \cos \frac{\theta}{2} |0\rangle + e^{i(\varphi+\gamma)} \sin \frac{\theta}{2} |1\rangle \\ &= e^{i\gamma} \left(\cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle \right) \end{aligned} \quad (2.2)$$

The overall phase factor does not have any observable effect, so it can be ignored. Hence, the state is described by:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle \quad (2.3)$$

A very visual way to illustrate the state of a qubit is by using angles θ and φ to represent a point in a three-dimensional sphere, known as the Bloch sphere, as shown in figure 2.1.

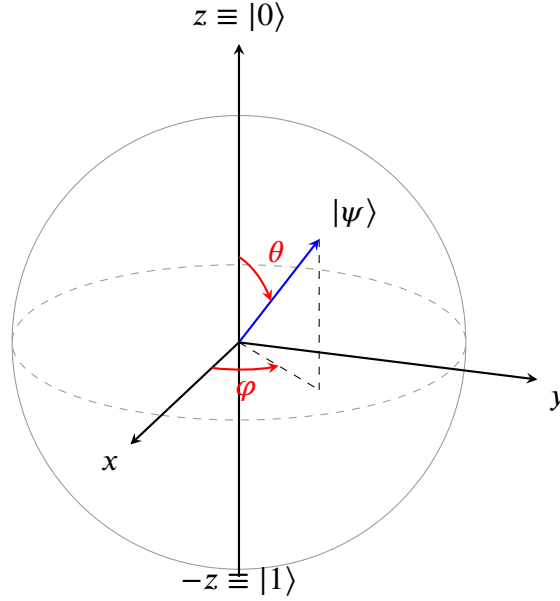


Figure 2.1: Bloch Sphere

The representation of more complex information requires several qubits. A state of N qubits lives in a vector space defined as the tensor product of the N two-dimensional Hilbert spaces corresponding to each individual qubits, producing a 2^N Hilbert space with a basis given by:

$$\begin{aligned}
 \underbrace{|0\rangle \otimes \cdots \otimes |0\rangle}_N &= \underbrace{|0 \cdots 00\rangle}_N \\
 \underbrace{|0\rangle \otimes \cdots \otimes |1\rangle}_N &= \underbrace{|0 \cdots 01\rangle}_N \\
 &\vdots \\
 \underbrace{|1\rangle \otimes \cdots \otimes |1\rangle}_N &= \underbrace{|1 \cdots 11\rangle}_N
 \end{aligned}$$

A quantum computer is composed by qubits and quantum gates, unitary operations which enable the manipulation of qubits. A computation can be depicted in the standard circuit model as the preparation of initial states, sequential application of quantum gates and measurements. Some quantum gates of interest are described in table 2.1.

A gate can be represented as a matrix, and its action on a quantum state is computed by multiplying said matrix with the column vector representation of the state. The total action of a circuit is obtained by the sequential execution of matrix multiplications.

Standard circuit model graphically represents each qubit as a horizontal wire, with quantum gates acting on them depicted as symbols on the corresponding wire. Time flows from left to right, thus determining the order in which the operations are executed. A simple example

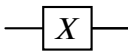
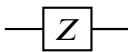
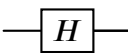
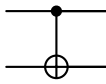
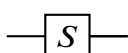
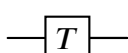
Gate Name	Symbol	Matrix representation
Pauli-X		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Z		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
CNOT		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
S Gate		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
T Gate		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$

Table 2.1: Relevant quantum gates.

of a quantum circuit can be seen in figure 2.2.

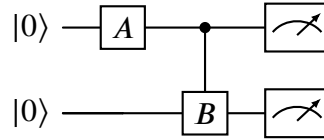


Figure 2.2: Example quantum circuit. Two qubits are prepared in the $|0\rangle$ state. Gate A acts on the first qubit. Then a controlled B gate is executed, where qubit 1 acts as control and qubit 2 acts as target. Finally, measurements in the computational basis are performed.

A set of gates is said to be universal if any unitary operation may be approximated to an arbitrary accuracy by some sequence of operations in this set.

3 | Quantum Error Correction

The main challenge of quantum computation is the fragility of the physical systems used to implement qubits. In classical computation, information is encoded in macroscopic physical variables, and the logical states are separated by energy barriers that make them robust against environmental fluctuations. Compared to classical bits, qubits are heavily affected by environmental noise, which makes them degrade and lose their information due to decoherence. It is thus necessary to explore the most efficient ways to protect quantum information while maintaining the practical feasibility of operating with those protected qubits. [35]

3.1 Overview and Simple Codes

Fathoming the differences between classical and quantum error-correcting codes makes it necessary to pay attention to three difficulties that arise when we deal with quantum systems: The no-cloning theorem, which prevents us from duplicating the state of a qubit; the continuous nature of errors; and the fact that observation in quantum mechanics collapses the wave function in one of the eigenstates of the measured observable.

In order to overcome these difficulties, quantum error-correcting codes (QECC) utilize entanglement to encode a logical qubit across a multi-qubit system. This way, we can protect quantum information through redundancy, without violating the no-cloning theorem. QEC protocols measure multi-qubit parity operators (error syndromes), avoiding the collapse of the wave function. The continuity problem will be confronted by discretizing the errors into a finite set of Pauli operations.

We will build some basic intuition about quantum error correction models based on redundancy by looking at one of its earliest and simplest examples: the Shor Nine-Qubit Code.

Consider that we have physical qubits that are subject to random bit-flip errors, that is, with a certain probability p , the state $|\psi\rangle$ is transformed into $X|\psi\rangle$. As we cannot simply duplicate our original qubit, we can entangle it with two ancilla qubits initialized in the $|0\rangle$ state by applying two CNOT gates, where the original qubit is the control and the ancilla qubits are the targets. If our original state is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the resulting state will be $|\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle$. The encoding establishes a logical basis with states:

$$\begin{aligned} |0_L\rangle &= |000\rangle \\ |1_L\rangle &= |111\rangle \end{aligned}$$

If the bit-flip error occurs in any qubit, we can diagnose this error without destroying the superposition by measuring the parity observables $Z_1 Z_2$ and $Z_2 Z_3$. Those observables compare the parities of their corresponding qubits, yielding a ± 1 eigenvalue, positive if both compared qubits share the same state, negative if they are different. It is crucial to note that the measurement of the parity operators commutes with the logical operators, ensuring that the superposition remains intact. Once we have detected the error, we can apply the appropriate recovery operation.

This same procedure can be adapted to protect against phase-flip errors, modeled by the Pauli Z operator. This is performed by applying a Hadamard transformation to each qubit after applying two sequential CNOT gates. It is an identical procedure, and it only differs in the basis we are working with.

If we concatenate the bit-flip and phase-flip codes, we can protect against an arbitrary error on a single qubit. This is therefore a nine-qubit code, with the base states defined as follows:

$$\begin{aligned} |0_L\rangle &= \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle) \\ |1_L\rangle &= \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle) \end{aligned}$$

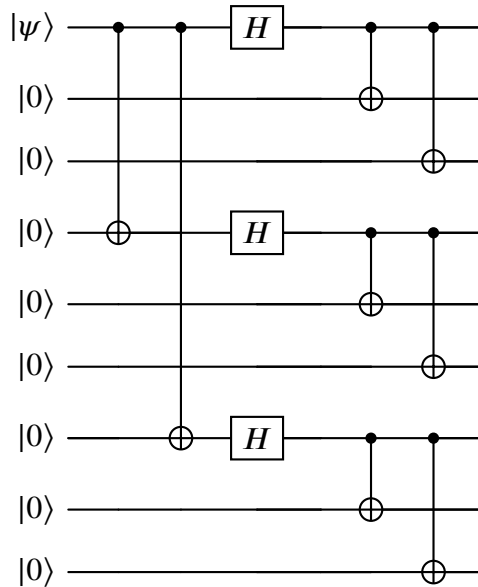


Figure 3.1: Shor Code circuit

The syndrome measurement is performed in an analogous way. We first compare the first two qubits by applying the operator $Z_1 Z_2$, then we continue comparing each pair of neighboring qubits by applying the corresponding operators. We can detect the bit-flip errors on

any of the qubits. In order to detect phase-flip errors, we compare the signs of each of the three-qubit blocks.

The Shor code allows us to detect bit-flip and phase-flip errors. In order to illustrate that this condition is enough to detect an arbitrary error, we will describe the noise by a trace-preserving operation $\mathcal{E}$. Given ρ the density matrix of the system, the noise acting on the state can be expressed as:

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger \quad (3.1)$$

This is called Kraus operator-sum representation, and will be discussed in more detail. The operators E_k can be expanded as a linear combination of the Pauli matrices $E_k = e_{i0}I + e_{i1}X_1 + e_{i2}Z_1 + e_{i3}X_1Z_1$. The resulting quantum state $E_k |\psi\rangle$ can be expressed as a superposition of $|\psi\rangle$, $X_1 |\psi\rangle$, $Z_1 |\psi\rangle$, and $X_1Z_1 |\psi\rangle$. When we measure the error syndrome, the system collapses to any of the four states, and after applying the necessary correction operation, we obtain the final state $|\Psi\rangle$.

3.2 General Theory of Quantum Error Correction

Once we have built some intuition about QEC, we can construct a more general theory.

Firstly, we need to give a mathematical description of our noise $\mathcal{E}$ [19]. Physically, noise represents an interaction between the system and the environment. Since we do not have access to the environment, the action of noise on the system appears to be a non-unitary operation, and we cannot describe our qubit as if it was in a pure state. This forces us to use the density matrix

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i| \quad (3.2)$$

that represents the mixture of different pure states $|\psi_i\rangle$ with probabilities p_i . This must be a positive semi-definite Hermitian matrix, with unit trace and non-negative eigenvalues. Non-unitary noise operations must preserve these properties. Hence, we represent noise acting on a system as a transformation of the state described by a density matrix that fulfills this condition: a Completely Positive Trace Preserving (CPTP) map. In the Kraus representation, the action of a CPTP map $\mathcal{E}$ on a density matrix ρ is given by the formula 3.1, where the Kraus operators E_k satisfy the condition $\sum_k E_k^\dagger E_k = I$ (this condition derives necessarily from the requirement of trace preservation). If we consider any state $|\psi\rangle$ in the Hilbert space and define the states $|\psi_k\rangle = E_k^\dagger |\psi\rangle$, we can easily check that:

$$\sum_k \langle \psi | E_k \rho E_k^\dagger | \psi \rangle = \sum_k \langle \psi_k | \rho | \psi_k \rangle \geq 0$$

which verifies that the map is positive. In fact, Choi's theorem on completely positive maps guarantees that any quantum operation admitting such a Kraus representation is inherently *completely positive*, meaning it yields valid physical states even when applied to a subsystem of a larger entangled system. We will call the operation elements $\{E_i\}$ errors, and they represent the transformation of the quantum state under Hamiltonian parameter deviations from their ideal values.

Now we want to generalize the basic ideas present in the Shor code to give a simple picture of how we can detect and correct errors in a quantum system. We firstly define a quantum error-correcting code as a subspace C of the total Hilbert space, and the elements of C will be called codewords. We encode our quantum states by projecting them into C by a unitary operation P that we will call the projector. The encoded system is then vulnerable to environmental noise, which we model as a CPTP map $\mathcal{E}$. We perform a syndrome measurement to determine the error that has affected the system, and a recovery operation $\mathcal{R}$ returns the system to its original encoded state.

When we project our state into the code subspace, we are essentially restricting the degrees of freedom of the system. This means that if our original Hilbert space for n physical qubits is 2^n -dimensional, the code subspace will be 2^k -dimensional, where $k < n$. We will call C a $[[n, k]]$ code, as we are encoding k logical qubits into n physical qubits. Once the noise acts on the encoded state $|\psi_L\rangle$, the system will be pushed into a different subspace. Specifically, each error E_α projects the state into an error subspace $C_\alpha = E_\alpha C$. Each error subspace has its corresponding projector $P_\alpha = E_\alpha P E_\alpha^\dagger$.

If, given a code, each error corresponds to different undeformed and orthogonal subspaces, then we can reliably distinguish which error has occurred. This will be achieved by performing the set of projective measurements associated with each error subspace. Given our noisy state $E_\alpha |\psi_L\rangle$, the execution of a projector P_β will yield us a syndrome reading β with probability $p_\beta = \langle \psi_L | E_\alpha^\dagger P_\beta E_\alpha | \psi_L \rangle$. The orthogonality of error subspaces ensures that the measurement associated to an error subspace C_β will only yield a non-zero probability when the corresponding error has occurred.

The undeformed condition means that, if you have two perfectly distinguishable (orthogonal) states $|\psi_L\rangle$ and $|\phi_L\rangle$ in the code subspace, then the error E_α must not make them non orthogonal. This is equivalent to saying that the inner product of the two states must be preserved under the action of the error, that is, $\langle \psi_L | E_\alpha^\dagger E_\alpha | \phi_L \rangle = \langle \psi_L | \phi_L \rangle = 0$. This ensures that the recovery operation can correctly restore the original encoded state.

We can define the condition for an error correction code to be able to correct the effect of noise as follows: Given a code C , a noisy quantum channel characterized by $\mathcal{E}$, and any state $|\psi_L\rangle$, with density matrix ρ_L , the error correction is successful if

$$(\mathcal{R} \circ \mathcal{E})(\rho) \propto \rho \quad (3.3)$$

In this condition, and in the following equation, we have not imposed the trace-preserving condition for $\mathcal{E}$, with the idea of allowing for more general noise models. That is the reason we

use the proportionality symbol $\propto$. The prior discussion can be summarized in a very compact algebraic expression, the Knill-Laflamme condition for quantum error correction [30]:

Theorem 3.1 (Knill-Laflamme). *Let C be a quantum error-correcting code, P the projector onto C , and $\mathcal{E}$ a CPTP map with operation elements $\{E_i\}$. Then, there exists a recovery operation $\mathcal{R}$ such that $(\mathcal{R} \circ \mathcal{E})(\rho) = \rho$ for all ρ supported on C if and only if the following condition holds:*

$$PE_i^\dagger E_j P = \alpha_{ij} P \quad (3.4)$$

for some Hermitian matrix α_{ij} .

Proof. ($\Leftarrow$) We notice that the Hermitian matrix α_{ij} can be diagonalized by a unitary transformation u , such that $u\alpha u^\dagger = d$, where d is a diagonal matrix. We can define new operation elements $F_i = \sum_j u_{ij} E_j$. Those can be proven to be operation elements of $\mathcal{E}$. The new operation elements also satisfy the Knill-Laflamme condition, since:

$$\begin{aligned} PF_i^\dagger F_j P &= \sum_{k,l} u_{ki}^\dagger u_{jl} P E_k^\dagger E_l P \\ &= \sum_{k,l} u_{ki}^\dagger u_{jl} \alpha_{kl} P \\ &= d_{kl} P \end{aligned}$$

which simplifies the Knill-Laflamme condition, since d_{kl} is a diagonal matrix. This diagonalization makes it explicit that the error subspaces are orthogonal. We decompose the operation elements as $F_i P = \sqrt{d_{ii}} U_i P$, where U_i is a unitary operator and $\sqrt{d_{ii}}$ is a positive real number.

F_i maps the code subspace C to the error subspace defined by the projector $P_i = U_i P U_i^\dagger = F_i P U_i^\dagger / \sqrt{d_{ii}}$. As we have already discussed, the syndrome measurement is the projective measurement defined by the projectors P_i . The recovery operation $\mathcal{R}$ is defined as $\mathcal{R}(\rho) = \sum_k U_k^\dagger P_k \rho P_k U_k$. For any state ρ supported on C , we can show that:

$$\begin{aligned} U_i^\dagger P_i F_k \sqrt{\rho} &= U_i^\dagger P_i^\dagger F_j P \sqrt{\rho} \\ &= \frac{U_i^\dagger U_i P F_i^\dagger F_j P \sqrt{\rho}}{\sqrt{d_{ii}}} \\ &= \delta_{ij} \sqrt{d_{ij}} \sqrt{\rho} \end{aligned}$$

Thus, one can easily check that this recovery operation satisfies the error correction condition:

$$\begin{aligned} (\mathcal{R} \circ \mathcal{E})(\rho) &= \sum_{i,j} U_i^\dagger P_i F_j \rho F_j^\dagger P_i U_i \\ &= \sum_{i,j} \delta_{ij} d_{ii} \rho \propto \rho \end{aligned}$$

($\Rightarrow$) Assume $\{E_i\}$ are errors correctable by $\mathcal{R}$, with operation elements $\{R_j\}$. We define a quantum operation $\mathcal{E}_C$ by $\mathcal{E}_C(\rho) = \mathcal{E}(P\rho P)$. $P\rho P \in C$ for any ρ , so we can apply the error correction condition to obtain:

$$\mathcal{R}(\mathcal{E}_C(\rho)) \propto P\rho P$$

We can rewrite the left-hand side of this equation as:

$$\mathcal{R}(\mathcal{E}_C(\rho)) = \sum_{i,j} R_j E_i P \rho P E_i^\dagger R_j^\dagger$$

Thus

$$\sum_{i,j} R_j E_i P \rho P E_i^\dagger R_j^\dagger = c P \rho P$$

for some constant c . The quantum operation with operation elements $\{R_j E_i\}$ is identical to the quantum operation with only one operation element $\sqrt{c}P$. This implies that there exist a complex matrix with entries c_{ki} such that $R_k E_i P = c_{ki} P$. We take the adjoint and obtain $P E_i^\dagger R_k^\dagger R_k E_j P = c_{ki}^* c_{kj} P$. Since $\mathcal{R}$ is a CPTP map, we have $\sum_k R_k^\dagger R_k = I$. Thus, we can sum over k to obtain:

$$P E_i^\dagger E_j P = \alpha_{ij} P$$

Which is the Knill-Laflamme condition, with $\alpha_{ij} = \sum_k c_{ki}^* c_{kj}$.

|

The proof of the Knill-Laflamme condition explicitly provides a method to construct the recovery operation $\mathcal{R}$, given the code subspace and the noise model. However, a quantum system could be affected by a noise process that is not known. It would be desirable to have a code C and an error-correction operation $\mathcal{R}$ that can correct the effect of an entire class of noise processes. That is indeed possible, and it is a consequence of the linearity of quantum mechanics. If we have a code that can correct the effect of a set of errors $\{E_i\}$, then it can also correct the effect of any error that can be expressed as a linear combination of those errors. This is a direct consequence of theorem 3.1, since if C satisfies the Knill-Laflamme condition for $\{E_i\}$, then it also satisfies it for any linear combination of those errors.

3.3 Stabilizer Codes

The previous discussion has proven the theoretical possibility of quantum error correction. However, the state-vector representation of quantum states carries a practical impossibility.

When we have a large number n of physical qubits, the manipulation of the 2^n -dimensional state vector becomes unfeasible. We need a more efficient way to represent quantum states, and the stabilizer formalism provides us with such a tool, as well as a method to construct QECC.

3.3.1 Stabilizer Formalism

If we take a closer look at the Shor code, and specifically at the syndrome measurement, we can see that in order to detect a bit flip on one of the first three qubits, we measured the eigenvalues of the operators $Z_1 Z_2$ and $Z_2 Z_3$. The same procedure should be performed for the other two blocks of three qubits. In order to detect phase-flip errors, we need to measure the eigenvalues of the operators $X_1 X_2 X_3 X_4 X_5 X_6$ and $X_4 X_5 X_6 X_7 X_8 X_9$. In total, we need to measure the eigenvalues of eight operators. The two valid codewords, $|0_L\rangle$ and $|1_L\rangle$, are the eigenstates of all these operators with eigenvalue $+1$, and hence are simultaneously fixed by any of them. Those are called stabilizer operators, and the basic idea of the formalism is to describe the code space as the subspace stabilized by these operators.

In order to give a more precise description of the stabilizer formalism, let's introduce the n -dimensional Pauli group $\mathcal{P}_n$, which is the group generated by the n -fold tensor products of the Pauli matrices and the identity, together with the multiplicative factors ± 1 and $\pm i$. For instance, the 1-dimensional Pauli group is

$$\mathcal{P}_1 = \{\pm I, \pm X, \pm Y, \pm Z, \pm iI, \pm iX, \pm iY, \pm iZ\}$$

It is easily verifiable that $\mathcal{P}_n$ is a group under matrix multiplication. The stabilizer is some Abelian subgroup $S \subseteq \mathcal{P}_n$ that does not contain $-I$, and the vectors fixed by all elements of S form the code space $C(S)$, i.e.:

$$C(S) \equiv \{|\psi\rangle \in \mathcal{H} : g|\psi\rangle = |\psi\rangle; \forall g \in S\}$$

It is also obvious that the code space $C(S)$ is indeed a vector space. S is necessarily an Abelian subgroup, since two non-commuting operators cannot have common eigenstates.

Let's recall that our aim is to define a subspace C of the total Hilbert space, such that it encodes k logical qubits into n physical qubits. Therefore, the dimension of C must be 2^k . We want to see how this condition translates on the definition of the stabilizer S :

Proposition 3.1. If S is an Abelian subgroup of $\mathcal{P}_n$ that does not contain $-I$, then the code space $C(S)$ stabilized by S has dimension 2^{n-k} , where k is the number of independent generators of S .

Proof. Let S be generated by $\langle g_1, g_2, \dots, g_k \rangle$. Since all the generators are elements of $\mathcal{P}_n$, they are unitary and Hermitian, and thus have eigenvalues ± 1 . We can project the total Hilbert

space into the +1 eigenspace of each generator g_i by applying the projector

$$P_i = \frac{I + g_i}{2}$$

The code space $C(S)$ is the subspace stabilized by all the generators, and thus can be obtained by applying the projectors associated with each generator to the total Hilbert space:

$$P_C = \prod_{i=1}^k P_i = \prod_{i=1}^k \frac{I + g_i}{2}$$

If we expand the product, we sum over all possible products of the generators g_i , which yields the 2^k elements of S .

$$P_C = \frac{1}{2^k} \sum_{g \in S} g$$

The dimension of the code space $C(S)$ is given by the trace of the projector P_C :

$$\dim(C(S)) = \text{Tr}(P_C) = \frac{1}{2^k} \sum_{g \in S} \text{Tr}(g)$$

We have the following cases for the trace of an element $g \in S$:

- If $g = I^{\otimes n}$, then $\text{Tr}(g) = 2^n$.
- If $g \neq I^{\otimes n}$, then g is a non-trivial element of $\mathcal{P}_n$, and thus has trace zero, since it contains at least one Pauli matrix that has zero trace.

$-I \notin S$ ensures that there are no elements of S with trace -2^n . Thus, the only element of S with non-zero trace is the identity, and we have:

$$\dim(C(S)) = \text{Tr}(P_C) = \frac{1}{2^k} \cdot 2^n = 2^{n-k}$$

|

Therefore, a code that encodes k logical qubits into n physical qubits can be defined by a stabilizer S with $n - k$ independent generators. This proposition also provides a reason for the non-inclusion of $-I$ in the stabilizer. This is nonetheless obvious from the fact that the only solution to $-I |\psi\rangle = |\psi\rangle$ is the trivial solution $|\psi\rangle = 0$.

3.3.2 Unitary Dynamics of Stabilizer Codes and the Gottesman-Knill Theorem

We have discussed the description of a subspace using stabilizer formalism. Now let's see how we can describe the dynamics of those subspaces under the action of noise and other quantum

processes. Let $|\psi\rangle$ be an element of a code space $C(S)$ stabilized by S . If we apply a unitary operation U ,

$$U|\psi\rangle = U g |\psi\rangle = (U g U^\dagger) U |\psi\rangle \quad (3.5)$$

and we see that $U|\psi\rangle$ is stabilized by the conjugated operator $U g U^\dagger$. Since this is common to any element of $C(S)$, we can easily see that $UC(S)$ is stabilized by $USU^\dagger \equiv \{U g U^\dagger : g \in S\}$. This property does not ensure that any stabilizer of a subspace $UC(S)$ is a valid stabilizer, since the conjugation by U might introduce elements that are not in $\mathcal{P}_n$. We hence introduce the Clifford group C_n as the normalizer of the Pauli group:

$$C_n \equiv \{U : U \mathcal{P}_n U^\dagger = \mathcal{P}_n\}$$

The Clifford group is the largest group of unitary operations that preserve the Pauli group under conjugation. If we apply an operation $U \in C_n$ to a code space C stabilized by S , the resulting subspace UC is stabilized by $USU^\dagger \in \mathcal{P}_n$. Conjugation by a Clifford operation also preserves the commutation relations

$$U g_1 U^\dagger U g_2 U^\dagger = U g_1 g_2 U^\dagger = U g_2 g_1 U^\dagger = U g_2 U^\dagger U g_1 U^\dagger$$

which ensures that the resulting stabilizer $USU^\dagger$ is also an Abelian subgroup of $\mathcal{P}_n$. If $-I$ was in $USU^\dagger$, then there would be some $g \in S$ such that $U g U^\dagger = -I$, and therefore:

$$\begin{aligned} U^\dagger (U g U^\dagger) U &= U^\dagger (-I) U \\ g &= -I \end{aligned}$$

which contradicts the definition of stabilizer. Thus, $-I$ is not in $USU^\dagger$, and the resulting subspace UC is a valid code space stabilized by $USU^\dagger$.

The Clifford group contains the Pauli group, but is larger. C_n is generated by the Hadamard gate, the phase gate, and the CNOT gate. It is important to note that this small set is enough to create highly entangled states.

This formalism provides us with a method to efficiently represent the dynamics of quantum circuits under the action of Clifford operations. The idea is that, instead of representing the state of the system as a vector in a 2^n -dimensional Hilbert space, we can represent it by its stabilizer, which is generated by n independent generators. Each generator is an element of $\mathcal{P}_n$, and thus can be represented by a string of n characters, each of which can be I , X , Y , or Z . Therefore, we can represent the state of the system by a list of n strings of length n . This discussion culminates in the following theorem:

| Theorem 3.2 (Gottesman-Knill). *Any quantum circuit that consists solely of Clifford gates, preparation of qubits in the computational basis, and measurements of Pauli operators can be efficiently simulated on a classical computer.*

The Clifford group is not universal for quantum computation, and thus the Gottesman-Knill theorem does not contradict the fact that quantum computers can outperform classical

computers. However, it is important to note that the Clifford group is a very large group of operations, and it is enough to create highly entangled states, and many quantum error correction circuits, as well as quantum protocols such as teleportation or superdense coding can be implemented using only Clifford gates. In order to achieve universality, we need to add at least one non-Clifford gate, such as the T gate.

3.3.3 Error Detection and Correction in Stabilizer Codes

Now that we have established the stabilizer formalism, we can explicitly describe how a stabilizer code detects and corrects errors in a remarkably elegant way. [20]

Let $C(S)$ be a $[n, k]$ stabilizer code defined by the stabilizer group $S = \langle g_1, g_2, \dots, g_{n-k} \rangle$. A valid encoded state $|\psi_L\rangle \in C(S)$ satisfies $g_i |\psi_L\rangle = |\psi_L\rangle$ for all $i \in \{1, \dots, n-k\}$.

Suppose an arbitrary error occurs. As discussed in Section 3.1, we can discretize errors by expanding them in the basis of the Pauli group $\mathcal{P}_n$. Thus, we only need to consider what happens when a Pauli error $E \in \mathcal{P}_n$ corrupts our state, transforming it into $E |\psi_L\rangle$. To detect the error without measuring the logical state (which would destroy the quantum information), we measure the $n-k$ stabilizer generators g_i .

Since both E and g_i belong to the Pauli group $\mathcal{P}_n$, they must either commute ($Eg_i = g_iE$) or anti-commute ($Eg_i = -g_iE$). This fundamental property of the Pauli group dictates the outcome of our syndrome measurements:

- If E commutes with g_i : $g_i(E |\psi_L\rangle) = Eg_i |\psi_L\rangle = E |\psi_L\rangle$. The corrupted state is an eigenstate of g_i with eigenvalue $+1$.
- If E anti-commutes with g_i : $g_i(E |\psi_L\rangle) = -Eg_i |\psi_L\rangle = -E |\psi_L\rangle$. The corrupted state is an eigenstate of g_i with eigenvalue -1 .

The outcomes of these $n-k$ measurements form the *error syndrome*. Typically, we map the eigenvalues ± 1 to binary values $s_i \in \{0, 1\}$, where $s_i = 0$ corresponds to $+1$ (commutes) and $s_i = 1$ corresponds to -1 (anti-commutes). The syndrome vector $\vec{s} = (s_1, s_2, \dots, s_{n-k})$ provides a unique fingerprint for the error E , provided that the code is capable of correcting it.

Once the syndrome $\vec{s}$ is extracted, classical processing (decoding) is used to determine the most likely error E that occurred. The correction step simply consists of applying a recovery operator $R \in \mathcal{P}_n$ that shares the same syndrome as E . If the decoding is successful, RE will commute with all stabilizers, and ideally $RE \in S$. Since elements of S act as the identity on the code space, the combined effect of the error and the recovery operation is trivial, successfully restoring the original state: $RE |\psi_L\rangle = |\psi_L\rangle$.

However, not all errors can be successfully corrected. If an error E commutes with all stabilizer generators but is not an element of S , it will yield a trivial syndrome (all $+1$) while

transforming the state into a different, orthogonal codeword. The set of all operators in $\mathcal{P}_n$ that commute with all elements of S is the centralizer of S , which for the Pauli group is equivalent to the normalizer $N(S)$.

Therefore, undetectable and uncorrectable logical errors are exactly those in $N(S) \setminus S$. The minimum weight (defined as the number of non-identity tensor factors in the Pauli string) of an operator in $N(S) \setminus S$ defines the distance d of the code. Such a code is usually denoted by the parameters $[n, k, d]$, meaning it encodes k logical qubits into n physical qubits and can correct any arbitrary error affecting up to $\lfloor (d - 1)/2 \rfloor$ physical qubits.

We can build a deeper structural understanding of the code by considering the logical characterisation of errors instead of just tracking them as a raw collection of physical errors. This means that we can find different elements of $N(S) \setminus S$ that act on the encoded states in the same way. Specifically, for any given error $E \in N(S) \setminus S$, we can find an element $g \in S$ such that $Eg \in N(S) \setminus S$, and $Eg |\psi_L\rangle = E |\psi_L\rangle$ for any $|\psi_L\rangle \in C(S)$. This means that we could build a group of logical operators where different physical errors differing only by elements of S are identified in the same class of equivalence. This is obviously the definition of the quotient group $N(S)/S$. The elements of this group are also the valid logical operators of the code, as they are the only operators that act non-trivially on the encoded states without pushing them out of the code space.

3.4 Fault-Tolerant Quantum Computation

We have discussed how encoding quantum information into a subspace can protect the data from environmental noise. However, the very process of encoding, syndrome measurement, as well as the application of logical quantum gates can also be affected by noise. The fundamental idea of fault-tolerant quantum computation is to design these processes in such a way that they do not propagate errors uncontrollably, and that the error correction procedure can still successfully correct the errors that occur during these operations.

Multi-qubit gates carry the risk of propagating errors from one physical qubit to another. The reiterative application of such operations can produce a single physical hardware failure to cascade into a large number of errors on the physical qubits encoding the logical state. If the weight of the resulting error exceeds the distance d of the code, the error correction procedure will fail. Therefore, the design of fault-tolerant quantum circuits must ensure that errors do not propagate in a way that exceeds the error-correcting capabilities of the code.

We can prevent catastrophic error propagation by imposing the condition of transversality on the implementation of logical gates. A transversal gate is one that can be implemented by applying independent physical operations exclusively between corresponding physical qubits in different code blocks. This architectural method ensures that a single physical error propagates to at most one physical qubit in each code block, thus maintaining the error weight within the correctable threshold of the code. This may seem like a very simple condition, but it has a profound limitation [17]:

Theorem 3.3 (Eastin-Knill). *No local quantum error-correcting code can implement a universal set of logical gates transversally.*

This theorem implies that the implementation of transversal gates alone does not suffice to achieve universal fault-tolerant quantum computation. For the QECC we will be interested, the set of gates that can be implemented transversally are exactly the Clifford gates [8]

If we want universal quantum computation with transversal gates, the barrier imposed by the Eastin-Knill theorem can be surpassed through magic state injection [7], which enables the transversal execution of a T gate (Clifford+ T is a universal set of quantum gates). This is performed by the preparation and injection of a magic state $|T\rangle$ which, via the execution of Clifford gates, consumes this resource and executes a T gate on the data qubit. For the preparation of said states, the method of magic state distillation is used. The method consists of the "purification" of a certain number of copies of a mixed state, and, although we are not interested now in the specific realisation of the method, we should note that it is an expensive procedure in terms of its computational cost.

3.5 CSS Codes

A classical linear code C $[n, k]$ is a vector subspace of dimension k inside the binary vector space $\mathbb{Z}_2^n$. It is generated by a $k \times n$ matrix G whose rows are a basis in C . They codify a logical message $x \in \mathbb{Z}_2^k$ in a codeword $v \in \mathbb{Z}_2^n$ via the operation $v = Gx$. An equivalent way of defining the linear code is with the parity check matrices, which provides a more useful description in terms of error correction. A $[n, k]$ code is defined as the subspace containing all vectors x in $\mathbb{Z}_2^n$ such that $Hx = 0$, where H is an $n - k$ by n matrix known as the parity check matrix, with entries in $\mathbb{Z}_2$. Given a code C , there exists a dual code $C^\perp$, given by

$$C^\perp = \{w \in \mathbb{Z}_2^n \mid v \cdot w^\perp = 0, \forall v \in C\}$$

The construction of a CSS (Calderbank-Shor-Steane) code [1] requires two classical linear codes C_1 (a $[n, k_1]$ code) and C_2 (a $[n, k_2]$ code), which must satisfy $C_2 \subset C_1$. The dimension of the quotient subspace C_1/C_2 is $k_1 - k_2$. This defines the dimension of our code subspace, as its elements will be constructed as the following coherent superposition:

$$|x + C_2\rangle = \frac{1}{|C_2|} \sum_{y \in C_2} |x + y\rangle \quad (3.6)$$

where x is an element of C_1 . Suppressing the global phase, any operator in $\mathcal{P}_n$ can be mapped to a binary vector of $2n$ components $(u|v) \in \mathbb{Z}_2^{2n}$, where $u \in \mathbb{Z}_2^n$ codifies the support of the X matrices and $v \in \mathbb{Z}_2^n$ codifies the support of the Z matrices. Given a state $|z\rangle$ defined by the vector $z \in \mathbb{Z}_2^N$, a stabilizer $Z(u)$ defined by $u \in \mathbb{Z}_2^N$ acts on $|z\rangle$ as

$$Z(u) |z\rangle = (-1)^{(u \cdot z)} |z\rangle$$

For $|z\rangle$ to be stabilized by $Z(u)$, the exponent must be an even number and hence $u \cdot z = 0 \pmod{2}$. A set of Z stabilizers form a matrix H_Z such that any state z stabilized by it must

satisfy $H_Z z = 0$, so it works as a parity check matrix. Now consider a state $|x\rangle$ defined by the vector $x \in \mathbb{Z}_2^n$, the stabilizer $X(v)$ defined by $v \in \mathbb{Z}_2^n$ acts on $|x\rangle$ as

$$X(v) |x\rangle = |x + v\rangle$$

A state stabilized by $X(v)$ must be a coherent superposition of states $|x\rangle + |v\rangle$. Therefore, if we construct a matrix H_X such that every row of the matrix stabilizes elements of a certain subspace C_2 , this subspace is constituted by coherent superpositions of vectors $|x\rangle \in \mathbb{Z}_2^n$ and rows v of H_X . Now we recall that S must be abelian. The commutation rule $[X(u), Z(v)] = 0$ requires that

$$H_X H_Z^T = 0 \tag{3.7}$$

which implies that any row g_i in H_X satisfies $H_Z g_i^T = 0$. So, if C_1 is the subspace given by the parity check matrix H_Z and C_2 is the subspace given by the generator matrix H_X , $C_2 \subset C_1$, so for any two elements $z \in C_1$ and $x \in C_2$, $z+x$ is also an element C_1 . From the discussion above, the subspace simultaneously stabilized by the rows of H_X and H_Z is equivalent to C_1/C_2 . CSS codes can then be seen as a family of stabilizer codes where the stabilizer is generated only by X and Z operators.

4 | Topological Quantum Error Correction

We have introduced the algebraic framework for diagnosing and correcting quantum errors. However, the practical implementation of QECC must face another critical challenge: the strict near-neighbor connectivity of physical qubits in real scalable hardware platforms, such as superconducting circuits [3] [21]. This constraint necessitates the development of quantum error-correcting codes that can be implemented with local interactions, i.e., families of stabilizer codes where the number of qubits in the support of the stabilizer generators must be bounded. This will be the departure point of topological quantum codes, a concept initiated by Kitaev's toric code [29].

4.1 Homology

We are interested in a specific family of stabilizer codes, namely surface codes. Those can be regarded as encoding quantum information in the homological degrees of freedom of a two-dimensional lattice. In order to understand the construction of surface codes, we need to introduce some concepts from algebraic topology [5].

Let's consider a connected, closed, orientable surface of genus g , and define a lattice on it, with V vertices, E edges and F faces. We will call 0-cells the vertices of the lattice, 1-cells the edges, and 2-cells the faces.

We want to obtain the homology groups of this surface over $\mathbb{Z}_2$. For this purpose, we construct an abelian group for each dimension $i \in \{0, 1, 2\}$, generated by the sets of i -cells, where the field of coefficients is $\mathbb{Z}_2$. In this sense, if we consider a set of 1-cells $E = \{e_1, e_2, \dots, e_m\}$, we can represent it as a sum

$$c = \sum_i c_i e_i \quad c_i = \begin{cases} 0 & e_i \notin E \\ 1 & e_i \in E \end{cases}. \quad (4.1)$$

Such sums are called 1-chain. We can consider addition of 1-chains (taking into account that the coefficients are in $\mathbb{Z}_2$ and hence $e_i + e_i = 0$), obtaining an abelian group structure on the set of 1-chains C_1 . The elements of C_1 can be considered as binary vectors of length

E , and topologically $C_1 \simeq \mathbb{Z}_2^E$. Equivalently, we define the group of 0-chains $C_0 \simeq \mathbb{Z}_2^V$ as the abelian group generated by the sets of 0-cells, and the group of 2-chains $C_2 \simeq \mathbb{Z}_2^F$ using the exact same procedure for 2-cells.

The entire homology theory rests on the definition of the boundary operators $\partial_i : C_i \rightarrow C_{i-1}$, which are a family of group homomorphisms that take an i -chain and return its geometric boundary as an $(i - 1)$ -chain. Given our construction, we are interested in two boundary operators:

- $\partial_1 : C_1 \rightarrow C_0$ takes a 1-chain (a set of edges) and returns the set of vertices that are the endpoints of an odd number of edges in the 1-chain. For instance, if we have a 1-chain $c = e_1 + e_2 + e_3$, where e_1 connects vertices v_1 and v_2 , e_2 connects vertices v_2 and v_3 , and e_3 connects vertices v_3 and v_4 , then $\partial_1(c) = v_1 + v_4$, since v_1 and v_4 are the endpoints of an odd number of edges in c , while v_2 and v_3 are the endpoints of an even number of edges in c .
- $\partial_2 : C_2 \rightarrow C_1$ takes a 2-chain (a set of faces) and returns the set of edges that are the endpoints of an odd number of faces in the 2-chain.

The boundary operators satisfy the property that they are nilpotent of degree 2, meaning that $\text{im}(\partial_{i+1}) \subseteq \ker(\partial_i)$, or equivalently $\partial_i \circ \partial_{i+1} = 0$. This property is a direct consequence of the geometric interpretation of the boundary operators, since the boundary of a boundary is always empty.

From these boundary operators, we construct two subgroups of C_i . The group of i -cycles Z_i is defined as the kernel of ∂_i , that is, the set of i -chains that have an empty boundary. The group of i -boundaries B_i is defined as the image of ∂_{i+1} , that is, the set of i -chains that are the boundary of some $(i + 1)$ -chain. Since $\partial_i \circ \partial_{i+1} = 0$, we have that $B_i \subseteq Z_i$.

We can define the i -th homology group as the algebraic quotient group of i -cycles by i -boundaries:

$$H_i = \frac{Z_i}{B_i} \quad (4.2)$$

where the quotient group is defined as the set of equivalence classes of i -cycles, where two i -cycles are equivalent if they differ by a i -boundary, that is, the elements of H_i are of the form $[z] = \{z + b : b \in B_i\}$, where z is any i -cycle. The homology groups are topological invariants, meaning that they do not depend on the specific lattice we have defined on the surface, but only on the topology of the surface itself. For a connected, closed, orientable surface of genus g , we have that $H_0 \simeq \mathbb{Z}_2$, $H_1 \simeq \mathbb{Z}_2^{2g}$ and $H_2 \simeq \mathbb{Z}_2$.

4.2 Surface Codes

Surface codes are a type of CSS codes, where the X and Z stabilizers have their support on neighbouring sets of qubits. Consider a lattice embedded in a closed surface, and attach a qubit

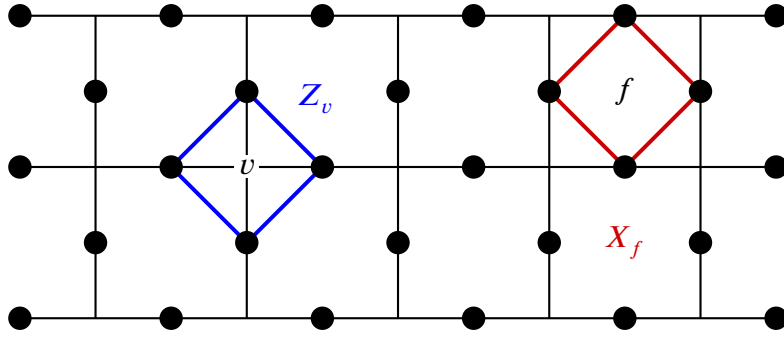


Figure 4.1: Lattice showing the arrangement of physical qubits and the support of the stabilizer generators.

to each edge. It is natural to see elements of the computational basis as 1-chains $c \in C_1$

$$|c\rangle = \bigotimes_i |c_i\rangle \quad (4.3)$$

It is useful to define groups homomorphism from C_1 to the group of Pauli operators $\mathcal{P}_n$, where n is the number of edges in the lattice and hence the number of physical qubits. We can define two homomorphisms, $X : C_1 \rightarrow \mathcal{P}_n$ and $Z : C_1 \rightarrow \mathcal{P}_n$, such that

$$X_c = \bigotimes_i X_i^{c_i} \quad \text{and} \quad Z_c = \bigotimes_i Z_i^{c_i}. \quad (4.4)$$

Given these homomorphisms, we can define the face (or plaquette) and vertex (or star) operators as follows:

$$X_f := \prod_{e \in \partial_2 f} X_e \quad \text{and} \quad Z_v := \prod_{e \in \partial_1 v} Z_e \quad (4.5)$$

Those operators are illustrated in figure 4.1, giving a clear intuition of their geometric meaning. The face and vertex operators commute with each other, and hence we can consider a stabilizer group $\mathcal{S}$ generated by them. In order to see the code subspace stabilized by $\mathcal{S}$, we can apply the projectors associated with each generator to any state of the total Hilbert space $|c\rangle, c \in C_1$:

$$\prod_f \frac{I + X_f}{2} \prod_v \frac{I + Z_v}{2} |c\rangle$$

Let's look at how each of the projectors acts on our arbitrary 1-chain state $|c\rangle$. The vertex operator Z_v effectively checks the parity of the edges of the chain c connected to the vertex v :

$$Z_v |c\rangle = (-1)^k |c\rangle \quad \text{where } k = |\{e \mid v \in \partial_1 e, e \in c\}|$$

From the definition of the boundary operator ∂_1 , we can see that $Z_v |c\rangle = |c\rangle$ if and only if v is the endpoint of an even number of edges in c and hence $v \notin \partial_1 c$. If $v \in \partial_1 c$, then $Z_v |c\rangle = -|c\rangle$ and the projector $(I + Z_v) |c\rangle = 0$. This means that the only states that survive the application of all vertex projectors are those associated with 1-cycles, $|z\rangle$, with $z \in Z_1$.

Now we apply the face projectors to our surviving states $|z\rangle$:

$$\prod_f \frac{1 + X_f}{2} |z\rangle$$

We can expand the product of all face terms and obtain a sum over all possible subsets of faces $\{f_i\}$. We also use that applying face operators on adjacent faces cancels out the operators on their shared edges:

$$\prod_f (1 + X_f) = \sum_{\{f_i\}} \prod_i X_{\partial_2 f_i} = \sum_{\{f_i\}} X_{\partial_2(\sum_i f_i)}$$

We notice that $\sum_i f_i$ is a 2-chain, and thus $\partial_2(\sum_i f_i)$ is a 1-boundary. Therefore this is proportional to the sum over all possible 1-boundaries:

$$\prod_f (1 + X_f) \propto \sum_{b \in B_1} X_b$$

Applying this to our cycle state $|z\rangle$ yields:

$$\prod_f \frac{1 + X_f}{2} |z\rangle \propto \sum_{b \in B_1} X_b |z\rangle = \sum_{b \in B_1} |z + b\rangle \quad (4.6)$$

This means that the states associated with two different cycles differing by a boundary are projected to the same state, and hence the code subspace stabilized by $\mathcal{S}$ is the vector space generated by the states $|[z]\rangle = \sum_{b \in B_1} |z + b\rangle$, where $[z]$ is the equivalence class of z in H_1 . Since $H_1 \simeq \mathbb{Z}_2^{2g}$, we have that the code subspace has dimension 2^{2g} , and hence encodes $k = 2g$ logical qubits into n physical qubits.

Now, let's consider the effects of bit-flip and phase-flip errors on the code subspace. We can model bit-flip errors as the application of X operators on the elements of a certain 1-chain c , so that its action on the physical state is given by $X_c |b\rangle = |b + c\rangle$, and on the encoded state $X_c |[z]\rangle = |[z + c]\rangle$. If the error chain is an open string ($\partial_1 c \neq 0$), then the error kicks the encoded state out of the protected subspace, so it can be easily detected. This means that X_z can map the code back to itself only if $z \in Z_1$. On the other hand, if the error acts on a 1-boundary $b \in B_1$, then the encoded state remains the same as the homology class does not

change $X_b | [z] \rangle = | [z] \rangle$. We conclude that the distance of a surface code for bit-flip errors is the length of the shortest non-bounding cycle $z \in Z_1 \setminus B_1$.

In order to proceed with the analysis of phase-flip errors, we need to construct the dual lattice. We will map faces to dual vertices f^* , edges to dual edges e^* (they represent the same physical qubits) and vertices to dual faces v^* . We have the corresponding dual operators $\partial_1^* : C_0^* \rightarrow C_1^*$ and $\partial_2^* : C_1^* \rightarrow C_2^*$, and they produce the groups of dual cycles Z_1^* , dual boundaries B_1^* and the homology group H_1^* . We can map the original code subspace to the one defined by the surface code on the dual lattice by applying a Hadamard gate $H^{\otimes E}$ to all physical qubits, giving us the following stabilizers:

$$H^{\otimes E} X_f H^{\otimes E} = \prod_{e \in \partial_2 f} Z_e = \prod_{e^* | f^* \in \partial_2^* e^*} Z_e =: Z_{f^*} \quad (4.7)$$

$$H^{\otimes E} Z_v H^{\otimes E} = \prod_{e | v \in \partial_1 e} X_e = \prod_{e^* \in \partial_1^* v^*} X_e =: X_{v^*}. \quad (4.8)$$

We can then treat phase-flip errors as bit-flip errors on the dual lattice, and applying the same reasoning as before, we conclude that the distance of a surface code for phase-flip errors is the length of the shortest non-bounding cycle $z^* \in Z_1^* \setminus B_1^*$ on the dual lattice.

If we recall the discussion on error detection in stabilizer codes, we can establish a clear parallelism between the normalizer and stabilizer groups, and the different 1-chain groups defined on the lattice. Firstly, we notice that we can write any element of the Pauli group $\mathcal{P}_E$ in terms of a pair of chains c, c^* , and a phase, as follows:

$$A = i^k X_c Z_{c^*} \quad (4.9)$$

where $k \in \{0, 1, 2, 3\}$ is an integer that depends on the specific element of $\mathcal{P}_E$. We want to find the elements of the normalizer, that is, operators A that commute with all the vertex and face operators. As we have already discussed, the commutation relations are:

$$[X_c, S_v] = 0 \iff v \notin \partial_1 c \quad [Z_{c^*}, S_f] = 0 \iff f \notin \partial_2^* c^* \quad (4.10)$$

This means that the normalizer is given by the set of operators $A = i^k X_c Z_{c^*}$ such that $\partial_1 c = 0$ and $\partial_2^* c^* = 0$, which is equivalent to saying that $c \in Z_1$ and $c^* \in Z_1^*$. The stabilizer is given by the subset of the normalizer where $c \in B_1$ and $c^* \in B_1^*$. Therefore, the logical operators are given by the elements of the normalizer that are not in the stabilizer, which correspond to pairs of cycles that are not boundaries. This is exactly the definition of the homology groups H_1 and H_1^* , and hence we have that the logical operators of a surface code are in one-to-one correspondence with the homology classes of the lattice and its dual:

$$N(S)/S \simeq H_1 \times H_1^* \quad (4.11)$$

4.3 Surface Codes with Boundaries

Embedding a lattice on a closed surface provides a mathematically elegant way to encode quantum information using local interactions. However, the practical feasibility of arranging

physical qubits on a closed surface such as a torus is questionable, but arbitrarily truncating a closed surface code to a planar geometry destroys the non-trivial homology classes. Bravyi and Kitaev [6] proposed a solution to this problem by introducing the concept of boundaries in surface codes. By carefully designing the boundaries of the lattice, we can preserve the non-trivial homology classes and hence the logical qubits, while still maintaining local interactions between physical qubits. This leads to the well-known planar surface code, which is one of the most promising candidates for practical quantum error correction in near-term quantum devices.

This extension of closed surface codes is based on a notion of relative homology [32], as presented in [16]. We consider our 2-dimensional surface with a lattice defined by the graph $G = (V, E, F)$ containing two categories of boundaries: open (or rough) and closed (or smooth). An edge is a boundary if it is incident to a single face of G , and the endpoints of these edges are boundary vertices.

We specify a subset of boundary edges and declare it as open, denoted $\partial_O E$. If a vertex is incident to an edge $E \in \partial_O E$, we call it an open vertex $v \in \partial_O V$. The remainder of boundary edges and vertices are called closed, defining $\partial_C V$ and $\partial_C E$. Physical qubits are allocated exclusively in the interior (non-open) edges, defining the interior sets $\tilde{V} = V \setminus \partial_O V$ and $\tilde{E} = E \setminus \partial_O E$. This means that the vector spaces associated with the chain complex are restricted to the non-open elements, with 0-chains and 1-chains defined as $C_0 = \bigoplus_{v \in \tilde{V}} \mathbb{Z}_2 v$ and $C_1 = \bigoplus_{e \in \tilde{E}} \mathbb{Z}_2 e$, while the 2-chains remain the same $C_2 = \bigoplus_{f \in F} \mathbb{Z}_2 f$.

The boundary operators $\partial_2 : C_2 \rightarrow C_1$ and $\partial_1 : C_1 \rightarrow C_0$ are defined as in the closed surface but restricting the image of both operators to the interior subsets. The truncation does not alter the nilpotency condition $\partial_1 \circ \partial_2 = 0$. Therefore we can define the group of relative 1-cycles Z_1^∂ as the kernel of ∂_1 and the group of relative 1-boundaries B_1^∂ as the image of ∂_2 , and $B_1^\partial \subseteq Z_1^\partial$. By discarding the open vertices in C_0 , the definition of a relative cycle now includes subsets of non-open edges with endpoints on open vertices, thus eliminating the necessity of forming a closed path. The first relative homology group is:

$$H_1^\partial = \frac{Z_1^\partial}{B_1^\partial} \quad (4.12)$$

Intuitively, the introduction of homological degrees of freedom seems obvious, as the boundaries introduce non-trivial cycles represented as open paths on the lattice. To determine the exact number of logical qubits k encoded by a surface code with boundaries, one needs to calculate the dimension of H_1^∂ using the rank-nullity theorem:

$$\dim H_1^\partial = \dim Z_1^\partial - \dim B_1^\partial \quad (4.13)$$

Proposition 4.1. The dimension of the first homology group of a generalized surface code is given by:

$$\dim H_1^\partial = -|\tilde{V}| + |\tilde{E}| - |F| + \kappa_{\partial_O V}(G) + \kappa_{\partial_C E}(G) \quad (4.14)$$

Where $\kappa_{\partial_o V}(G)$ denotes the number of connected components of G containing no open vertices and $\kappa_{\partial_c E}(G)$ denotes the number of connected components of G containing no closed edges.

Proof. Departing from (4.13), we first obtain $\dim Z_1^\partial$, the relative cycle space of the open graph $H = (V, E)$. We construct a 2-cover graph H_2 devoid of open vertices by taking two identical copies of H and connecting the corresponding pairs of open vertices with an extra edge. For this standard graph without open vertices, the dimension is given by $|E| - |V| + \kappa$, where κ is the number of connected components. Considering H_2 :

$$\dim Z_1(H_2) = 2|\tilde{E}| + |\partial_o V| - 2|\tilde{V}| + 2\kappa_{\partial_o V}(H) + \kappa_{\partial_o V}(H)$$

We define the linear projection $\pi : Z_1(H_2) \rightarrow Z_1^\partial$ that maps every cycle in H_2 onto its restriction in the first copy of H . The kernel of π is the set of cycles contained entirely in the second copy of H , and hence:

$$\dim \ker \pi = |\tilde{E}| - |\tilde{V}| + \kappa(\tilde{H})$$

And using the rank-nullity theorem:

$$\dim Z_1^\partial = \dim Z_1(H_2) - \dim \ker \pi = |\tilde{E}| - |\tilde{V}| + \kappa_{\partial_o V}(G) \quad (4.15)$$

Now for the relative boundaries, B_1^∂ is generated by the vectors $\partial_2 f$ for all $f \in F$. If a connected component C of G contains at least one closed boundary edge $e \in \partial_c E$, we cannot find a combination of its faces $c_f \in C_2$, with $c_i \neq 0$ only for $f_i \in C$, such that $\partial_2(c_f) = 0$, so the vectors $\partial_2 f$ associated with faces in this component are linearly independent. If the connected component lacks any closed boundary edge, summing the boundary operators of all faces in the component yields the zero vector. We then conclude that:

$$\dim B_1^\partial = |F| - \kappa_{\partial_c E}(G) \quad (4.16)$$

Thus, using equations (4.13), (4.15) and (4.16), we obtain:

$$\dim H_1^\partial = -|\tilde{V}| + |\tilde{E}| - |F| + \kappa_{\partial_o V}(G) + \kappa_{\partial_c E}(G)$$

|

There are different methods to encode a logical qubit using mixed boundaries, but we are primarily interested in continuous 2-dimensional planar patches completely devoid of internal holes or topological defects. Those are finite planar lattices forming a topological disk.

Consider that we assign a uniform boundary to our planar patch (either closed or open). It is immediate that

$$\kappa_{\partial_o V}(G) + \kappa_{\partial_c E}(G) = 1$$

as our lattice is simply connected and it has either open or closed boundaries, but not both. Applying the Euler characteristic for a disk:

$$|V| - |E| + |F| = |\tilde{V}| + |\partial_o V| - |\tilde{E}| - |\partial_o E| + |F| = 1$$

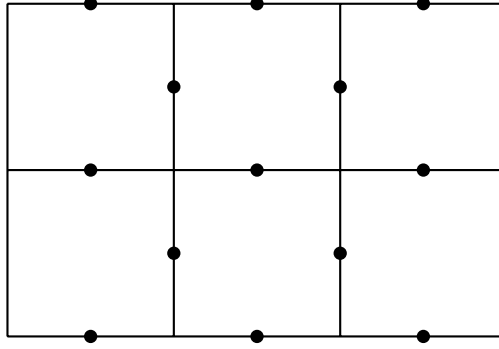


Figure 4.2: A planar patch. The top and bottom edges are closed (smooth) boundaries and the left and right edges are open (rough) boundaries. The black circles are once again physical qubits, and as we can see the open edges are not associated with any physical qubit.

If the entire boundary of the disk is closed, the number of both open vertices and open edges is 0. On the other hand, if the entire boundary is open, the number of vertices and edges in the perimeter are equal and hence $|\partial_o V| - |\partial_o E| = 0$. Therefore:

$$-|\tilde{V}| + |\tilde{E}| - |F| = -1$$

and using proposition 4.1:

$$\dim H_1^\partial = 0$$

Hence, we conclude the necessity of using mixed boundaries to encode a logical qubit on our planar patch. We specifically propose a rectangular planar surface code where opposing parts of the perimeter are assigned the same type of boundaries, as represented in figure 4.2.

For any of the sides of the perimeter, $|V_s| - |E_s| = 1$. Therefore, $|\partial_o V| - |\partial_o E| = 2$. Applying this and the Euler characteristic for the disk:

$$\begin{aligned} -|\tilde{V}| + |\tilde{E}| - |F| &= -|V| + |E| - |F| + |\partial_o V| - |\partial_o E| \\ &= -1 + 2 = 1 \end{aligned}$$

We also notice that $\kappa_{\partial_o V}(G) = \kappa_{\partial_c E}(G) = 0$. Therefore, the dimension of the first homology group $\dim H_1^\partial = 1$, successfully encoding one logical qubit in our planar surface code.

If we consider this prototypical realisation of the planar surface code, we see that Z-type star stabilizers at each vertex and X-type plaquette stabilizers at each face in the interior of the lattice simultaneously stabilize four qubits. On the other hand, vertices in the smooth boundaries are only connected to 3 edges, and therefore the Z-type stabilizers along the rough boundaries are weight-3 operators. If we consider faces adjacent to rough boundaries, they are bounded by only three interior edges (therefore three physical qubits), so the X-type plaquette stabilizers along the rough boundaries are weight-3 operators too. Those weight-3 stabilizers can be seen as algorithmic "sinks", allowing error chains to terminate on the boundaries of the lattice without pushing the system out of the code subspace.

4.4 Logical Operators and the Distance of the Code

Following the connection between homology and logical operators established for the closed surface, we can now define the logical operators for the planar surface code. The logical Pauli operators correspond to the non-trivial equivalence classes of the relative homology group H_1^∂ and its dual. A logical X_L operator is represented by a chain of physical X operators along a path connecting two distinct open (rough) boundaries. Such a path is a relative 1-cycle because its boundary endpoints lie entirely within the open vertices, meaning it has an empty boundary in the interior subset $\tilde{V}$. Hence, it commutes with all vertex stabilizers.

Similarly, by considering the dual lattice, a logical Z_L operator is formed by a chain of physical Z operators connecting two closed (smooth) boundaries. These operators commute with all stabilizers but cannot be constructed as a product of them, as they are not relative boundaries. As we showed, a standard planar surface code encodes a single logical qubit ($k = 1$) by featuring two opposite open boundaries and two opposite closed boundaries. This topology allows for one independent X_L path and one independent Z_L path. These paths must intersect at exactly one physical qubit (or an odd number of them), ensuring the necessary anti-commutation relation $\{X_L, Z_L\} = 0$ for the logical Pauli algebra.

The distance of the planar surface code is determined by the minimum weight of a non-trivial logical operator. For the X_L operator, this is the length of the shortest path between the two rough boundaries, often denoted as the horizontal width d_x . For the Z_L operator, the distance is the length of the shortest path between the two smooth boundaries, denoted as the vertical height d_z . The overall distance d of the code is $\min(d_x, d_z)$.

Executing the planar surface code demands continuous syndrome extraction across the lattice. In order to measure the stabilizers, ancillary measure qubits are interlaced among the data qubits [18], as shown in figure 4.3a.

Syndrome qubits are placed in the centre of each plaquette and in each vertex, and they measure the stabilizer by the application of a standardised sequence of CNOT gates, where the data qubits are control and the ancilla qubit is target, thus entangling the ancilla qubit with its surrounding data qubits. The circuits are shown in figures 4.3c and 4.3b.

In practice, we avoid the application of correcting physical operations once we detect errors, as they could introduce more noise in the system. Instead, we keep track of the accumulation of errors by use of a classical registry known as the Pauli frame [4]. The positive Pauli frame is that in which the encoded state is a +1 eigenstate of all the stabilizers.

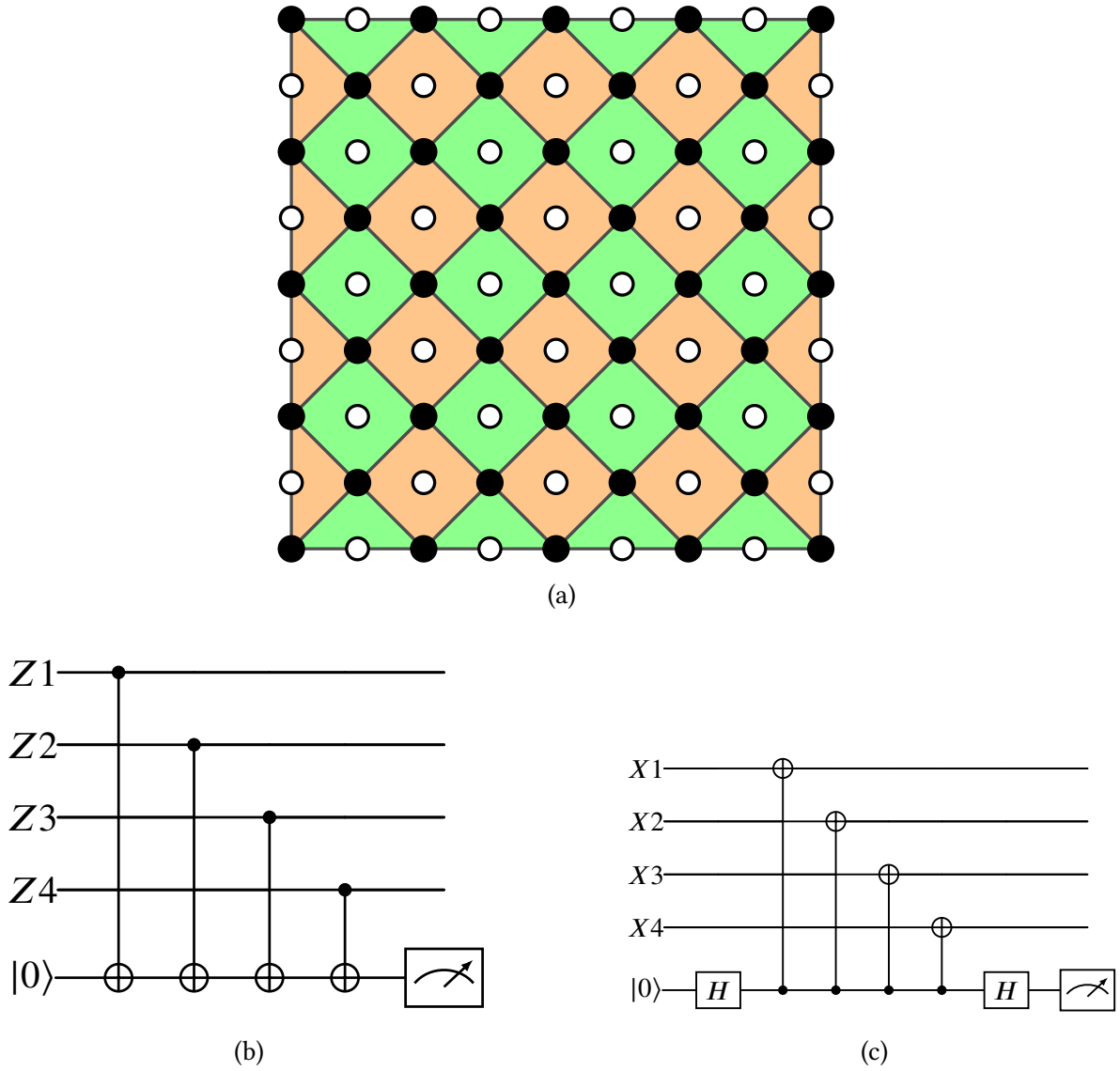


Figure 4.3: (a) A distance 5 planar surface code. Black dots represent data qubits, and white dots represent ancillary qubits. Green squares are X-type plaquettes stabilizers and orange squares are Z-type star stabilizers. (b) Z-syndrome. (c) X-syndrome.

5 | Lattice Surgery

Surface codes provide a promising method to encode logical qubits in fault-tolerant architectures. However, the standard transversal realisation of multi-qubit gates, where the physical qubits of different planar patches should interact with each other, would violate the 2-Dimensional Nearest-Neighbour (2DNN) connectivity. Lattice surgery constitutes a very useful approach, as it performs multi-qubit operations exclusively interacting the boundaries of the patches. Let's construct this approach using certain tools of category theory, as introduced in [14] and [15].

In order to understand the construction of lattice surgery, we need to review some basic concepts of category theory [13].

Definition 5.1. (i) Given a category $\mathcal{A}$ and an index category $\mathcal{I}$, a functor $D : \mathcal{I} \rightarrow \mathcal{A}$ is a diagram of shape $\mathcal{I}$ in $\mathcal{A}$.
(ii) A cone on D is an object $A \in \mathcal{A}$ and a family of morphisms $\psi_I : A \rightarrow D(I)$ for every $I \in \mathcal{I}$ such that, for every morphism $u : I \rightarrow J$ in $\mathcal{I}$, the triangle

$$\begin{array}{ccc} & D(I) & \\ \psi_I \nearrow & \downarrow D(u) & \searrow \\ A & & \\ \psi_J \searrow & \downarrow & \\ & D(J) & \end{array}$$

commutes.

(iii) A limit of D is a cone (L, π_i) over D such that, for any other cone (N, ψ_i) over D , there exists a unique morphism $h : N \rightarrow L$ such that, for all $I \in \mathcal{I}$, $\pi_i \circ h = \psi_i$.

In a more intuitive description, a diagram is a functor "stamping" the abstract structure of a simple category $\mathcal{I}$ in the richness of another category $\mathcal{A}$. The cone represents the commutative relations of a given object $A \in \mathcal{A}$ with a diagram D . For a certain diagram D , there could be several cones. The limit could be interpreted as the minimal object in $\mathcal{A}$ satisfying the commutation relations of a cone. We can define the dual concept of cone and limit.

Definition 5.2. (i) Given a diagram $D : \mathcal{I} \rightarrow \mathcal{A}$. A cocone on D is an object $A \in \mathcal{A}$ and a family of morphisms $\phi_I : D(I) \rightarrow A$ for every $I \in \mathcal{I}$ such that, for every morphism $u : I \rightarrow J$ in $\mathcal{I}$, the triangle

$$\begin{array}{ccc} D(I) & & \\ \downarrow D(u) & \searrow \phi_I & \\ & & A \\ & \nearrow \phi_J & \\ D(J) & & \end{array}$$

commutes.

(ii) A colimit of D is a cocone (L, π_i) over D such that, for any other cocone (N, ϕ_i) over D , there is a unique morphism $h : L \rightarrow N$ such that, for all $I \in \mathcal{I}$, $h \circ \pi_i = \phi_i$.

The mathematical substrate upon which we will operate is the category $\mathbf{Mat}_{\mathbb{F}_2}$. Its objects are finite-dimensional vector spaces over the field $\mathbb{F}_2$ and its morphisms are linear transformations. From this, we can define a structurally richer category: $\mathbf{Ch}(\mathbf{Mat}_{\mathbb{F}_2})$, the category of chain complexes. Its objects C are sequences of vector spaces over $\mathbb{F}_2$ connected by boundary morphisms ∂_n subject to the nilpotency condition:

$$\cdots \rightarrow C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \rightarrow \cdots \quad ; \quad \partial_n \circ \partial_{n+1} = 0 \quad (5.1)$$

This is nothing but the general definition from category theory of the structure we used to construct the surface codes. The morphisms in this category are chain maps $f : C \rightarrow D$ constituted by a family of linear transformations $f_n : C_n \rightarrow D_n$ that preserve the homological structure. This requires the following commutative square diagram:

$$\begin{array}{ccc} C_n & \xrightarrow{\partial_n^C} & C_{n-1} \\ f_n \downarrow & & \downarrow f_{n-1} \\ D_n & \xrightarrow{\partial_n^D} & D_{n-1} \end{array}$$

$\mathbf{Ch}(\mathbf{Mat}_{\mathbb{F}_2})$ is an abelian category, which ensures the existence of all its finite limits and colimits [13]. This allows to evaluate certain colimits that operate on chain complexes:

- **The Coproduct:** It is the colimit of a diagram $D : \mathcal{I} \rightarrow \mathcal{A}$ such that $\mathcal{I}$ only has two objects and no non-identity morphisms. In additive categories, like $\mathbf{Ch}(\mathbf{Mat}_{\mathbb{F}_2})$, the categorical coproduct and product are the same as the direct sum. The coproduct of objects A and B is denoted $A \amalg B$.

- **The Coequalizer:** It is the colimit of a diagram in $\mathcal{A}$ consisting of two objects, A and B , and two parallel morphisms f and g :

$$A \begin{array}{c} \xrightarrow{f} \\ \rightrightarrows \\ \xrightarrow{g} \end{array} B$$

Given the colimit morphism $\pi_B : B \rightarrow E$, where E is the coequalizer, the definition 5.2 imposes $\pi_B \circ f = \pi_B \circ g$. This means that the coequalizer is constructed using an equivalence relation between the images of f and g .

- **Pushout** It is the colimit of a diagram in $\mathcal{A}$ called a *span*, which consists of three objects A , B and C and two morphism f and g :

$$B \xleftarrow{f} A \xrightarrow{g} C$$

where A could be seen as a common object and f and g as inclusion maps. A pushout consists of an object P and two morphisms $i_1 : B \rightarrow P$ and $i_2 : C \rightarrow P$, such that $i_1 \circ f = i_2 \circ g$. We can construct the pushout using the coproduct and the coequalizer. We first take the coproduct of B and C , $B \amalg C$, with the canonical inclusion morphisms $i_B : B \rightarrow B \amalg C$ and $i_C : C \rightarrow B \amalg C$. The coproduct is too big, as it contains all of B and C , so the "shared" part A is represented twice. We need to map A into the coproduct, either mapping A into B and then into the coproduct, via $i_B \circ f$, or mapping A into C and then into the coproduct, via $i_C \circ g$. We have the diagram that defines the coequalizer:

$$A \begin{array}{c} \xrightarrow{i_B \circ f} \\ \rightrightarrows \\ \xrightarrow{i_C \circ g} \end{array} B \amalg C$$

Therefore, the colimit of this diagram gives us an object P , with its universal morphism $\pi : B \amalg C \rightarrow P$. The object P is the pushout.

Now consider that we have two chain complexes A and B in $\mathbf{Ch}(\mathbf{Mat}_{\mathbb{F}_2})$. If we want to merge those two spaces, we can define the surface along which they will be merged. In the context of planar codes, the boundary subcomplexes, constituted as subcomplexes of the boundary spaces of the original chain complexes ($S \subset \partial A$ and $S \subset \partial B$). As the boundary operators are still defined in S , the homology $H(S)$ is well defined. We define chain monomorphisms, that is, injective maps that commute with the boundary operators $f_{n-1} \circ \partial_n^S = \partial_n^A \circ f_n$, denoted as $f : S \rightarrow A$ and $g : S \rightarrow B$. This constitutes a span, $A \xleftarrow{f} S \xrightarrow{g} B$. The colimit of such a diagram, the pushout, can be computed for each degree, using the procedure we gave above, and considering that $\mathbf{Ch}(\mathbf{Mat}_{\mathbb{F}_2})$ is an abelian category and therefore the coproduct can be expressed as a direct sum:

$$M_n = \frac{A_n \oplus B_n}{\{f_n(s) \smile g_n(s) \mid s \in S_n\}} \quad (5.2)$$

As f and g are monomorphisms, there is a short exact sequence (the image of each map coincides with the kernel of the next one):

$$0 \rightarrow S \xrightarrow{(f,g)} A \oplus B \xrightarrow{\pi} M \rightarrow 0 \quad (5.3)$$

and, by the Snake Lemma [15], we have an exact long sequence:

$$\dots \rightarrow H_1(S) \xrightarrow{i_*} H_1^\partial(A) \oplus H_1^\partial(B) \xrightarrow{\pi_*} H_1^\partial(M) \xrightarrow{\delta} H_0(S) \rightarrow \dots \quad (5.4)$$

Where i_* is induced by (f, g) and π_* by π , while δ is the connector homomorphism given by the Snake Lemma. As it is an exact sequence, $\text{im}(i_*) = \ker(\pi_*)$. Therefore, any non-trivial cycle in S collapses to the trivial homological class in the merged complex M . This means that a logical operator along the merged boundary S becomes a trivial cycle and hence a stabilizer. $\text{im}(\pi_*) = \ker(\delta)$ means that the valid cycles in the merged space inherited from the original patches must have no boundaries contained in S . Intuitively, this is seen as the condition that logical operators originally ending at the merging face must concatenated to form a valid logical operator in the resulting surface.

In an standard merge operation, S is chosen in such a way that it codifies one logical operator, so $\dim H_1(S) = 1$. We also assume that its injection towards the original patches is non-trivial, so $\dim(\text{im} i_*) = 1$. As $\text{im}(i_*) = \ker(\pi_*)$, we know that, out of the $k_A + k_B$ original degrees of freedom present in the coproduct, a 1-dimensional subspace collapses towards the trivial class when projected by q_* towards M , so the dimension of the resulting code is $k_A + k_B - 1$. If the original patches encoded one logical qubit, the merged patch will also encode one logical qubit.

For the split operation, we must introduce a cut along the original patch M . The line along which we cut the manifold defines a chain subcomplex S . We can define two descending chain complexes, A and B , and construct a *cospan* diagram with two chain epimorphisms, $p : A \rightarrow S$ and $q : B \rightarrow S$ which constitute the following diagram:

$$B \xrightarrow{q} S \xleftarrow{p} A \quad (5.5)$$

The merged patch prior to the split is nothing but the limit of this diagram, also known as the pullback [14]. The abelian algebra guarantees that the pullback is evaluated as

$$M_n = \{(a, b) \in A_n \oplus B_n \mid p_n(a) = q_n(b)\} \quad (5.6)$$

The restriction forces that, prior to the split operation, any operator in region A that touches S must be perfectly continued by an operator in B . This split operation is defined as the application of the canonical morphisms of the limit $\pi_A : M \rightarrow A$ and $\pi_B : M \rightarrow B$. The system abandons M and is injected into the space $A \oplus B$. The kinematics of the logical information during the split operation is governed by the short exact sequence:

$$0 \rightarrow M \xrightarrow{i} A \oplus B \xrightarrow{p-q} S \rightarrow 0 \quad (5.7)$$

where i is the inclusion morphism defined by π_A and π_B . Once again, applying the Snake Lemma yields us a long exact sequence:

$$\dots \rightarrow H_1(M) \xrightarrow{i_*} H_1^\partial(A) \oplus H_1^\partial(B) \xrightarrow{p_*-q_*} H_1^\partial(S) \xrightarrow{\delta} H_0(M) \rightarrow \dots \quad (5.8)$$

The induced map i_* takes a logical cycle $[m] \in H_1(M)$ and maps it to the biproduct, yielding $([m_A], [m_B])$, representing the distribution of the original operator in the new two patches. As $\text{im}(i_*) = \ker(p_* - q_*)$, $p_*([m_A]) = q_*([m_B])$, which ensures that the restriction of the operators on the split surface S must be the same. If the split cuts through the original operator $[m]$, the restriction on S is not null, so the system prohibits the isolated existence of operator $[m_A]$ in the new space. This means that we cannot perform the measurement $[m_A] \otimes \mathbb{I}_B$, as it would

not commute with the stabilizers in the new structure. The categorical description of the split operation produces a notion of quantum entanglement.

By discussing the split and merge operations from the perspective of category theory, we have provided a solid mathematical background for said procedures, perfectly matching with the topological description of planar codes. However, discussing how those operations are performed by measuring observables on the boundaries of planar patches we will obtain a clearer intuition of the process, as well as a better understanding of the dynamics of the logical information encoded in the planar patches. We will also distinguish between smooth and rough operations, depending on which boundary we are using to perform the surgery.

5.1 Operational Description of Lattice Merging

Consider two planar patches A and B encoding one logical qubit each, stabilized by vertex X-type stabilizers and face Z-type stabilizers (what we called dual lattice in the previous sections will be considered as the original lattice now) [22]. The independent stabilization of said patches does not contradict the possibility of them sharing an open boundary, as the edges in said boundaries do not represent physical qubits (figure 4.2). We could consider this shared open (rough) boundary as a column of uninitialized physical qubits, that will constitute the support of rough merging. The two systems are merged by initiating the intermediate physical qubits in the state $|0\rangle$ and then performing d rounds of error correction, where d is the distance of the code. The arrangement is represented in figure 5.1

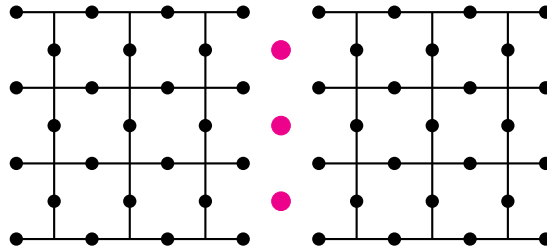


Figure 5.1: Physical distribution of qubits for a rough lattice merging operation. Black dots represent physical qubits subject to the stabilizers of their corresponding patch, and pink qubits are uninitialized physical qubits.

We determine the sign of the new X-stabilizers along the merged boundary, and therefore the eigenvalue of their product, which is equivalent to measuring the $X_{L,A} \otimes X_{L,B}$ operator. This measurement forces the collapse of the global state towards an eigenvalue of the operator, thus collapsing one degree of freedom, as the non-trivial chain representing $X_{L,A} \otimes X_{L,B}$ becomes now a trivial chain and hence a stabilizer. As the joint operator acts as the logical identity on the unified code subspace, the operators $X_{L,A}$ and $X_{L,B}$ are now equivalent, as they can be deformed from one patch to another by the new stabilizers. On the other hand, $Z_{L,A}$ and $Z_{L,B}$ which originally ended at the merged surface now have interior boundaries, and hence they stop being valid logical operators. A new logical operator Z_L is defined, constituted by the concatenation of both $Z_{L,A}$ and $Z_{L,B}$. This analysis perfectly matches the prior discussion

about the merging as a pushout in $\mathbf{Ch}(\mathbf{Mat}_{\mathbb{F}_2})$, where we concluded the collapse of a degree of freedom as well as of a valid logical operator in S , and also predicted the concatenation of logical operators with endpoints on S .

If we recover the construction of the dual lattice, a smooth merge can be defined, initializing the intermediate qubits in the states $|+\rangle$ and measuring the new joint operators, equivalent to measuring $Z_{L,A} \otimes Z_{L,B}$. The following discussion can be perfectly developed for a smooth merge, but it will be omitted to avoid redundancy.

The projective measurements performed during the merging operation necessarily produce the non-unitary evolution of the system, which is clearly manifested by the reduction of the dimensionality of the logical space [4]. The measurement of the operator $X_{L,A} \otimes X_{L,B}$ projects the global state towards one of its eigenspaces, revealing a global eigenvalue $M = \pm 1$. If $M = +1$ (the positive branch), the parity forces that in the merged patch $X_{L,A} \equiv X_{L,B}$. If $M = -1$ (the negative branch), the parity forces that in the merged patch $X_{L,A} \equiv -X_{L,B}$, as if one of the original patches suffered a chain of Z errors. The non-unitary evolution can be described by a CPTP map. In Kraus representation:

$$M_R(\rho) = K_{0,R}\rho K_{0,R}^\dagger + K_{1,R}\rho K_{1,R}^\dagger \quad (5.9)$$

Where each Kraus operator $K_{b,R}$ represents the evolution in the case of $M = (-1)^b$. When $M = 1$, there is no discrepancy and $K_{0,R}$ projects the correlation in the Hadamard base $K_{0,R} = |+\rangle_M \langle ++| + |-\rangle_M \langle --|$. For the negative branch, instead of applying a correcting Z chain, the asymmetry is absorbed in the Pauli frame by considering a Z error in either of the original patches. The selection of which patch will be treated as if it suffered an error will determine the Kraus operator for the negative branch. For instance, if we correct the right patch, $K_{1,R} = |+\rangle_M \langle +-| + |-\rangle_M \langle -+|$, forcing the second logical qubit to align with the sign of the first. Let's try to produce a truth table using the Kraus operators to track the evolution of the initial logical state. The possible states are:

$$\begin{aligned} |00\rangle_L &= \frac{1}{2} (|++\rangle_L + |+-\rangle_L + |-+\rangle_L + |--\rangle_L) \\ |01\rangle_L &= \frac{1}{2} (|++\rangle_L - |+-\rangle_L + |-+\rangle_L - |--\rangle_L) \\ |10\rangle_L &= \frac{1}{2} (|++\rangle_L + |+-\rangle_L - |-+\rangle_L - |--\rangle_L) \\ |11\rangle_L &= \frac{1}{2} (|++\rangle_L - |+-\rangle_L - |-+\rangle_L + |--\rangle_L) \end{aligned}$$

For the positive branch:

$$\begin{aligned}
K_{0,R} |00\rangle_L &= \frac{1}{2} (|+\rangle_L + |-\rangle_L) = \frac{1}{\sqrt{2}} |0\rangle_L \\
K_{0,R} |11\rangle_L &= \frac{1}{2} (|+\rangle_L + |-\rangle_L) = \frac{1}{\sqrt{2}} |0\rangle_L \\
K_{0,R} |01\rangle_L &= \frac{1}{2} (|+\rangle_L - |-\rangle_L) = \frac{1}{\sqrt{2}} |1\rangle_L \\
K_{0,R} |10\rangle_L &= \frac{1}{2} (|+\rangle_L - |-\rangle_L) = \frac{1}{\sqrt{2}} |1\rangle_L
\end{aligned}$$

And for the negative branch:

$$\begin{aligned}
K_{1,R} |00\rangle_L &= \frac{1}{2} (|+\rangle_L + |-\rangle_L) = \frac{1}{\sqrt{2}} |0\rangle_L \\
K_{1,R} |11\rangle_L &= \frac{1}{2} (-|+\rangle_L - |-\rangle_L) = -\frac{1}{\sqrt{2}} |0\rangle_L \\
K_{1,R} |01\rangle_L &= \frac{1}{2} (-|+\rangle_L + |-\rangle_L) = -\frac{1}{\sqrt{2}} |1\rangle_L \\
K_{1,R} |10\rangle_L &= \frac{1}{2} (|+\rangle_L - |-\rangle_L) = \frac{1}{\sqrt{2}} |1\rangle_L
\end{aligned}$$

The truth table obtained from the rough merge corresponds with an XOR:

In(1)	In(2)	Out
0	0	0
0	1	1
1	0	1
1	1	0

5.2 Operational Description of Lattice Splitting

Lattice splitting can be viewed as the opposite operation of lattice merging. By a set of measurements on a row of physical qubits, a surface is split in two separately stabilized surfaces. Let's consider a rough split as represented in figure 5.2.

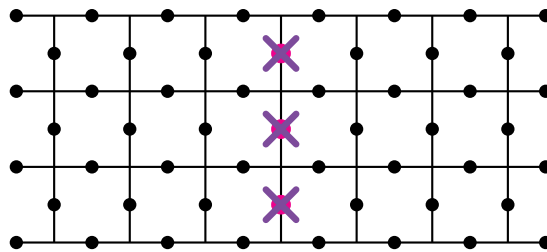


Figure 5.2: Physical distribution of qubits for a rough lattice splitting operation.

We perform Z measurements on the crossed qubits in 5.2, which conform an interrupted line connecting two smooth boundaries through a path of interior edges. This line of measurements annihilate the degrees of freedom of said qubits, effectively splitting the original surface M in two separate patches A and B . The Z -type stabilizers corresponding to the plaquettes adjacents to the measured qubits are truncated into boundary stabilizers of weight 3.

The logical operators $X_{L,A}$ and $X_{L,B}$ in the resulting patches commute with the measurements of the split operation, so $X_{L,A} \equiv X_{L,B} \equiv X_{L,M}$, thus dividing the original logical qubit in two entangled logical qubits. To understand why this does not violate the no-cloning theorem, we notice that the $Z_{L,M}$ operator also commutes with the measurements of the split operation, and hence $Z_{L,M} \equiv Z_{L,A} \otimes Z_{L,B}$, which implies a distribution of the Z_L operator, fixing the parity in the computational basis.

The probabilistic character of the Z measurements implies once again the ± 1 results. If all measurement outcomes are $+1$, then the separation is perfect and does not require any correction. However, if one measurement yields an eigenvalue -1 , the two truncated Z -type plaquette stabilizers adjacents to said measurement collapse towards their negative subspace. We update the Pauli frame and simulate a chain of X operators to correct the error. The rough split is naturally described by the Kraus operator $U_R = |++\rangle\langle +| + |--\rangle\langle -|$.

6 | ZX calculus

The ZX calculus is a graphical tool for the diagrammatic representation and manipulation of quantum computation, presenting an alternative to the standard quantum circuit model, firstly presented in [10]. We will give a brief introduction, as it will be a fundamental tool for the present work, but a complete review of ZX calculus can be consulted in [25]. A formal treatment as a monoidal category theory is presented in appendix A.

6.1 ZX diagrams

While standard circuit notation implements a rigid temporal sequence of unitary operations, ZX calculus abstracts quantum computations into flexible tensor networks, known as ZX diagrams. Structurally, those diagrams are composed by edges and specific nodes called spiders. The edges represent a two-dimensional index (a qubit) attached to the specific tensors represented by the spiders. Those tensors could represent operations, as well as projections and preparations.

The diagrams have two families of generators, defined with respect to the computational basis (Z-basis) and the Hadamard-transformed diagonal basis (X-basis). In this work, red nodes will represent the computational basis and green nodes the diagonal basis. When the edges are attached to a node, depending on which side they come from they will represent either input legs or output legs. We will use the standard convention where edges coming on the left side of the node correspond to input legs (or time flows from left to right). Nodes may have an associated angle, representing a relative phase. The translation between spiders and bra-ket notation can be summarized in the following equalities:

$$\begin{aligned}
 n \text{---} \text{green spider}(\alpha) \text{---} m &= \underbrace{|0 \dots 0\rangle}_m \underbrace{\langle 0 \dots 0|}_n + e^{i\alpha} \underbrace{|1 \dots 1\rangle}_m \underbrace{\langle 1 \dots 1|}_n \\
 n \text{---} \text{red spider}(\alpha) \text{---} m &= \underbrace{|+\dots+\rangle}_m \underbrace{\langle +\dots+|}_n + e^{i\alpha} \underbrace{|-\dots-\rangle}_m \underbrace{\langle -\dots-|}_n
 \end{aligned} \tag{6.1}$$

Therefore, each diagram can be associated with a linear map. Preparations and projections in the computational basis can be represented as:

$$\text{red spider}(\alpha) \text{---} = R_x(\alpha) |0\rangle \quad \text{---} \text{red spider}(\alpha) = \langle 0| R_x(\alpha) \tag{6.2}$$

The composition of those basic generators allow for the systematic construction of ZX diagrams. If an edge has an endpoint without a node, it is considered either an input or an output (depending on the convention we use for input and output legs). For example, a way to construct a CNOT gate with ZX calculus:

$$\begin{aligned}
 \text{CNOT} &= |0\rangle\langle 0| \otimes \langle 0| + \otimes |+\rangle\langle +| + |0\rangle\langle 0| \otimes \langle 0| - \otimes |-\rangle\langle -| \\
 &\quad + |1\rangle\langle 1| \otimes \langle 1| + \otimes |+\rangle\langle +| + |1\rangle\langle 1| \otimes \langle 1| - \otimes |-\rangle\langle -| \\
 &= \frac{1}{\sqrt{2}} |0\rangle\langle 0| \otimes I + \frac{1}{\sqrt{2}} |1\rangle\langle 1| \otimes X = \frac{1}{\sqrt{2}} \text{CNOT}.
 \end{aligned}$$

Hadamard gates in this work will be represented as blue lines connecting two nodes, as they are quite frequent and this allows us to avoid the saturation of the diagrams.

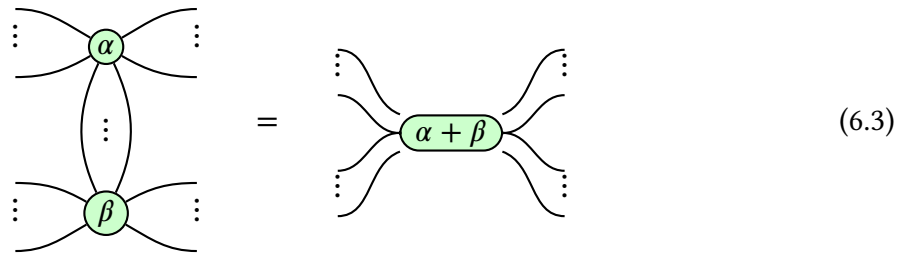
The deformation of a ZX diagram that do not alter its topology produces an isomorphic ZX diagram that describes the same linear map, given that the inputs and outputs are fixed [41]. This is commonly known as the meta-rule: only connectivity matters. Another fundamental property of ZX diagrams is what we call **universality**:

Theorem 6.1. Any linear map $L : \mathbb{C}^{2^n} \rightarrow \mathbb{C}^{2^m}$ can be represented as a ZX diagram [25].

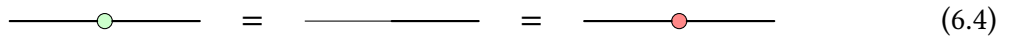
6.2 Rewrite Rules

A great advantage of ZX diagrams above plain tensor networks is that they have a collection of rewrite rules, which constitute ZX calculus in itself, and enable us to perform small changes in the diagram without altering the linear map associated. We will make a quick review of some of those rules. For the rigorous proofs of each rule, they are available in [45].

The most important rule in ZX calculus is **spider fusion**, that states that we can merge two spiders as long as they are of the same colour and connected by at least one edge:



The **identity removal** rule basically states that a rotation by 0 degrees does nothing, representing this rotation as a 1-input 1-output spider with phase :



A spider with a π phase can "cross" a spider with the opposite colour, "copying" its phase in each of the adjacent edges. This is called the **π -copy rule**:

$$\text{Diagram 1} = \text{Diagram 2} \quad (6.5)$$

As the X and Z basis are related by the Hadamard gate, the commutation of a Hadamard gate through a spider changes its color. This is the **colour change** rule:

$$\text{Diagram 1} = \text{Diagram 2} \quad (6.6)$$

When a "copy" structure of a certain colour interacts with a structure of parity evaluation of another colour, the graph can be rewritten using the **strong complementary** rule:

$$\text{Diagram 1} \propto \text{Diagram 2} \quad (6.7)$$

Although those rules may be enough for the reduction of Clifford ZX diagrams, we need to introduce more complex rewriting rules for Clifford+T ZX diagrams, represented in figure 6.1.

$$\begin{aligned} & \text{Diagram 1} \stackrel{(LC)}{=} e^{\pm i \frac{\pi}{4}} \sqrt{2}^{\frac{(n-1)(n-2)}{2}} \text{Diagram 2} \\ & \text{Diagram 3} \stackrel{(P)}{=} (-1)^{jk} \sqrt{2}^E \text{Diagram 4} \end{aligned}$$

Figure 6.1: The **local complementation** and **pivoting** rules. Figure from [41].

6.3 ZX calculus and lattice surgery

The non-unitary evolution of quantum systems under the operations of lattice surgery makes the standard circuit model unable to faithfully represent the dynamics of the procedures. On the other hand, ZX calculus has proven to be a natural way to represent lattice surgery, as described by [4]. We want to present this correspondence, founded in the mapping from the Kraus operators of lattice surgery to the spiders of ZX diagrams.

It is easy to see that the linear maps corresponding to 2-to-1 nodes in ZX diagrams perfectly match the Kraus operators for the positive branch of a lattice merge.

$$\text{green spider} = K_{0,S} \quad \text{red spider} = K_{0,R} \quad (6.8)$$

The correspondence between lattice split and 1-to-2 nodes is also immediate:

$$\text{green spider} = U_S \quad \text{red spider} = U_R \quad (6.9)$$

This immediately yields the result that the positive branches of lattice surgery procedures are naturally captured by ZX diagrams with nodes of degree 2 and 3. Nodes of greater degree, or including angles, can be described, using the spider fusion rule, as the composition of rotations and split and merge operations in the positive branch. If we consider the realisation of the CNOT by composing a rough merge and a smooth split, the positive branch is given by $(I \otimes K_{0,R})(U_S \otimes I) = \frac{1}{\sqrt{2}} \text{CNOT}$. Therefore, the subnormalisation we found when implementing the CNOT gate in ZX diagrams correspond to the fact that it only captures the positive branch of the procedure, which in this case has a probability of $1/2$.

For the negative branch of the lattice merge, we have to consider the adaptation of the Pauli frame, which is easily captured by ZX diagrams. It requires adding a π rotation prior to the 2-to-1 node. If we consider that the correction is applied on the second patch, we obtain:

$$\text{green spider with } \pi \text{ rotation} = K_{1,S} \quad \text{red spider with } \pi \text{ rotation} = K_{1,R} \quad (6.10)$$

For this work, we will restrict the study to the positive branch. This is physically justified as the dynamic of the negative branch does not alter the structure of the information processes, and it differs from the positive branch only by the action of additional Pauli operators. As the emergence of those Pauli operators is univocally heralded by the classical bit resulting from the merging measurement, they can be absorbed by the classical control system, avoiding the necessity of corrections in the quantum hardware. Therefore, when compiling the algorithm the hardware executes the same sequence of splits and merges, independently of the eigenvalues obtained.

It has been proven that the ZX calculus transcends its function as a mere graphical notation system, establishing itself as a native language of fault-tolerant quantum computation.

Its set of diagrammatic rewrite rules can be implemented for an automated optimization of quantum protocols.

To prove the utility of ZX calculus, let's use it to present a simple implementation of a T gate with lattice surgery. As we already discussed in chapter 3, magic states permit the realisation of non-Clifford operations on surface codes. A T gate specifically requires a magic state and teleportation to execute a $\pi/4$ rotation on a logical qubits. In [4], they propose an efficient execution of the T gate with lattice surgery operations. Restricting ourselves to the positive branch, we have:

$$\begin{array}{c} \pi/4 \\ \circ \\ \diagdown \\ \text{---} \circ \end{array} = \begin{array}{c} \pi/4 \\ \text{---} \circ \end{array} \quad (6.11)$$

The diagram in the left represents the preparation of a state $|A\rangle \propto |0\rangle = e^{i\pi/4}|1\rangle$, and merging said state with the target logical qubit, and the rules of ZX calculus prove it to be equivalent to a $\pi/4$ rotation.

7 | Practical Problem

The implementation of quantum algorithms in fault-tolerant architectures requires their treatment in a model that does not impose the unitary evolution of standard circuit models. The practical problem of this work can be summarized as the compilation of quantum protocols in terms of a problem of topological optimization of tensor networks using ZX calculus.

Given a quantum circuit $\mathcal{C}$, we have a translation map $\mathcal{T} : \mathcal{C} \rightarrow D_{ZX}$, where D_{ZX} is a ZX diagram obtained by the substitution of rotations in Z and X bases by green and red spiders, as well as the initialization of qubits and measurements [43].

The resource cost of a given computation is naturally captured by its space-time volume [46], defined as its spatial footprint (number of qubits) times its execution time (number of error correction cycles required). As lattice surgery operations exhibit a direct correspondence with spiders in ZX calculus, the topological deformation of a ZX diagram necessarily implies the restructuring of its physical operations. Hence, the compilation problem can be formulated as an optimization problem:

$$\min V(D'_{ZX}) \quad \text{such that} \quad [[D'_{ZX}]] = [[D_{ZX}]] \quad (7.1)$$

where $[[\cdot]]$ represents the lineal map described by the object. Applying rewrite rules allows to drastically optimize the metric. A quantitative evaluation of the space-time volume can be performed on ZX diagrams corresponding to lattice surgery procedures. The number of logical qubits required to implement a sequence of lattice surgery operations represented by a ZX diagram is given by the maximum number of wires in any vertical cut of the diagram [43].

For the temporal cost of simple protocols, we can use the layer slicing procedure [46]. We identify the entry nodes of the diagram (usually, the open wires in the left) and they are assigned the layer index $L = 1$. Their topological neighbours are assigned $L = 2$. We propagate this process, and each vertex is assigned a layer index $L(v) = \max_{u \in \text{fathers}(v)} L(u) + 1$. The temporal cost of an algorithm can be approximated by the maximum index in the diagram, which we will call the topological depth.

For simple diagrams, the computation of this metric is quite straightforward, but when the protocol requires a high number of quantum gates, the complexity of the diagrams hinders the computation of its topological depth. Multi-qubit gates such as the CNOT or CZ appear as vertical lines in the ZX diagrams, and the resolution of this ambiguity in one specific operational instantiation of the gate result in different graphs, hence altering the layer slicing procedure. Different physical realisations for the multi-qubit gates imply different topological depths. The

task of reducing the rounds of error correction rounds, and hence of improving the execution time of the algorithm can be seen as an optimization problem in graph theory [42]. We can try to formalize it by giving a more precise definition of topological depth.

Definition 7.1. *Let $G = (V, E)$ be a directed acyclic graph. We define a set of entry vertices as:*

$$V_1 = \{v \in V \mid \text{in-degree}(v) = 0\}$$

where the in-degree represents the number of edges directed towards v . For any vertex $u \in V \setminus V_1$, we define the depth or the layer index map $L : V \rightarrow \mathbb{N}$ in a recursive way:

$$L(u) = \max\{L(v) \mid (v, u) \in E\} + 1$$

where $(v, u) \in E$ in an edge directed from v to u . We define the topological depth of the graph as:

$$D = \max\{L(v) \mid v \in V\}$$

When our ZX diagram has differentiated wires representing different logical qubits, we can model it as a mixed graph $G = (V, E_H, E_V)$, where E_H are the horizontal edges along a wire and E_V are the vertical edges for each multi-qubit gate. The fact that the vertical edges admit different physical realisations can be translated to this set up by considering E_H as a set of directed edges such that they conform separate directed, non-intersecting, open paths (wires), and E_V as a set of non-directed edges connecting vertices in different horizontal paths. The direction of the edges can be associated with a time-like direction, and the first vertex of each wire represents an input of the diagram, and the last vertices correspond to the outputs. The specific physical realisation of the gate is an assignation of an orientation to its corresponding vertical edges. Given a mixed graph G , an orientation $\mathcal{O}$ is a function that assigns a direction to each $e \in E_V$. For an orientation $\mathcal{O}$ to be valid, the resulting graph $G^\mathcal{O} = (V, E_H \cup E_H^\mathcal{O})$ must be acyclic.

Furthermore, the physical impositions of the hardware allow us to restrict the graph model further. The graph contains a finite number of wires $\{W\}_i$. Given any wire W_1 , three vertices $u_1, u_2 \in W$ and two edges $(u_1, v_1), (u_2, v_2) \in E_V$ such that $v_1 \in W_2, v_2 \in W_3$ and $W_2 \neq W_3$, then if any other vertex $u_3 \in W_1$ has a vertical edge attached $(u_3, v_3) \in E_V$, then v_3 is either in W_2 or W_3 . The restriction introduces the concept of neighbour wires, i.e., wires in the graph that may be linked with vertical edges. Other conditions can be defined for different connectivity restrictions. This model is represented in figure 7.1. The depth of a vertex v can be interpreted as the longest directed path starting at any input vertex and ending at v .

Given this model for our diagrams, the time-optimization problem in a graph G is solved by searching the orientation $\mathcal{O}$ that yields the minimum topological depth $D(G^\mathcal{O})$. It is clear that $D(G^\mathcal{O}) \geq D(G)$, as the length of the directed paths cannot decrease after we apply an orientation.

In order to construct an algorithmic solution for the problem, let's consider how a specific iteration could be performed: we start reading the graph from the input vertices (in figure 7.1, from left to right) until we find the first vertex attached to a vertical edge e . The two vertices

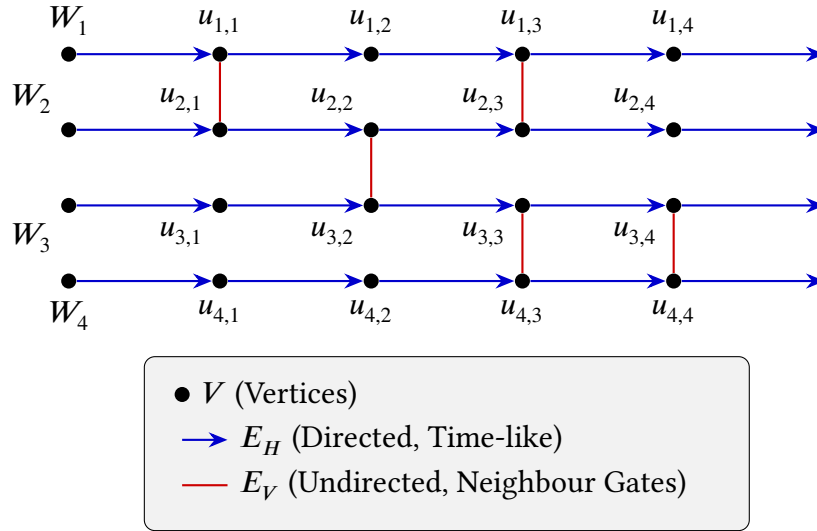


Figure 7.1

connected by this edge, v and u have a primitive layer index $L_{in}(u)$ and $L_{in}(v)$ respectively, determined by their preceding horizontal edges (and, for later iterations, by their preceding directed vertical edges too). If we direct e from $u \rightarrow v$, the new depth of v will be $L_{out}(v) = \max\{L_{in}(v), L_{in}(u) + 1\}$, and $L_{out}(u) = L_{in}(u)$. It is immediate that, if $L_{in}(v) > L_{in}(u)$, then $L_{out}v = L_{in}(u)$. Therefore, directing the edge towards the vertex with the highest depth does not produce any change on the depth of the vertices, and hence, on the total topological depth of the graph.

If we have a vertical edge connecting two vertices (u, v) with $L_{in}(v) = L_{in}(u)$, or strictly vertical edges, we need to define a new metric $S : V \rightarrow \mathbb{N}$, which we will call the shallowness of the vertex, and it will be given by the length of the horizontal path connecting the vertex in the argument and the output vertex of the wire. For an edge (u, v) with $L_{in}(v) = L_{in}(u) = k$ and, without loss of generality, $S(u) < S(v)$, let's consider the local impact of the orientation on the cost of their respective paths $L_{out}(v) + S(v)$ and $L_{out}(u) + S(u)$:

- $u \rightarrow v$: $L_{out}(u) = k$, and the total cost of its path will be $k + S(u)$. $L_{out}(v) = k + 1$ and the total cost will be $k + 1 + S(v)$. The maximum local cost is $\max\{k + S(u), k + 1 + S(v)\} = k + 1 + S(v)$
- $v \rightarrow u$: $L_{out}(v) = k$, and the total cost of its path will be $k + S(v)$. $L_{out}(u) = k + 1$ and the total cost will be $k + 1 + S(u)$. The maximum local cost is $\max\{k + S(v), k + 1 + S(u)\} = k + S(v)$

Hence, the orientation of the vertical edges should be towards the shallowest vertex. If we already assigned an orientation to every edge between vertices with different in-path depth (those do not produce any change in the depth, neither locally nor globally), we can take a global approach for a systematic treatment of the vertical edges. We have already proved that the orientation of those vertical edges produces an increase in the total length of one the wires. If we have several vertical edges connecting the two wires in vertices with the same in-path depth, once we direct the first one from the left, the remaining edges will have a trivial

orientation, as the depth of the vertices in one of the wires will increase by one. This means that if we direct all those vertical edges towards the same wire, the depth of the vertices in that wire will only increase by one. For instance, in figure 7.2, if we direct (u_1, v_1) and (u_2, v_2) upwards, $L_{out}(u_3) = L_{in}(u_3) + 1$. If a given wire W has those edges (between vertices with the same in-path depth) with two different wires, and we direct all of them towards W , the total depth of the wire will also increase by 1.

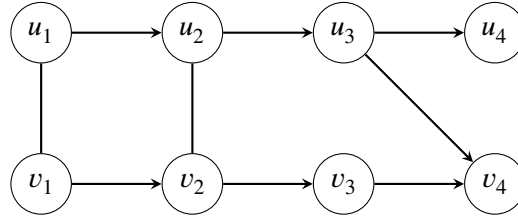


Figure 7.2

In cases where there are subsequent vertical edges connecting vertices u_i and v_j such that $L_{in}(v_j) - L_{in}(u_i) = 1$, (in figure 7.2, (u_3, v_4)) either orientation for the vertical edges will increase the depth of v_j , while if the vertical edges are directed towards the wire that contains v_j , then $L_{in}(u_i) = L_{out}(u_i)$. Therefore, in those situations the vertical edges should be directed towards the wire containing v_j .

If this case does not hold, a decision must be made, as one of the wires will necessarily suffer an increase in its depth. As we try to minimize the global depth of the graph, we make this decision by looking at the total length of all the wires. Firstly, we find the longest wire in the diagram, and direct all the strictly vertical edges attached to it towards its neighbours. This way we ensure that the total depth of the diagram does not increase. The remaining strictly vertical vertices will be oriented in an alternate way: if a wire already has strictly vertical edges oriented towards it (as in the case of the neighbours of the longest wire), the remaining strictly vertical edges attached to it will be also oriented towards it. The same thing applies for wires with directed vertical edges towards its neighbours. The criteria can be visualized in figure 7.3. If there are two wires with maximum length separated by an odd number of wires, this method does not present any conflict. If they are separated by an even number of wires, any of those intermediate wires would increase its total depth by 2, so the total depth of the diagram could increase by 1. Considering the case discussed above, that could also increase the length of the longest wire by 1, this method provides an orientation $\mathcal{O}$ such that the topological depth of the graph $D(G^{\mathcal{O}}) \leq D(G) + 2$.

The algorithm would be as follows:

- We assign the trivial orientations for the vertical edges connecting two vertices with different in-path depth.
- Among those oriented vertical edges, we search for edges (u_i, v_j) satisfying $L_{in}(v_j) - L_{in}(u_i) = 1$. If there are previous strictly vertical edges connecting the same two wires, those are directed towards the wire containing v_j .
- We identify the wires with the greatest length, and apply the assignation represented in 7.3

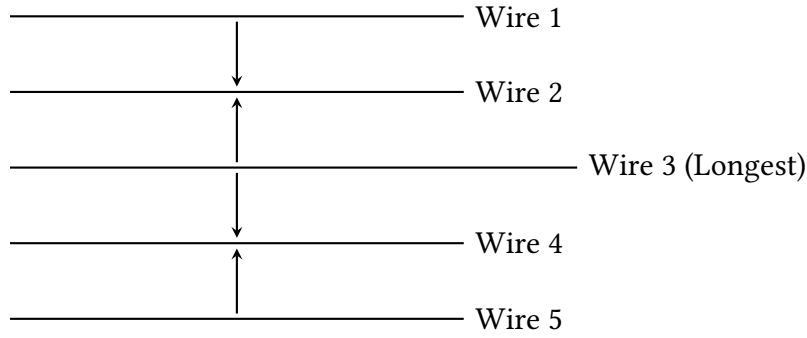


Figure 7.3

For the computational implementation of the problem, we will use PyZX framework, an open source python-based library specifically designed for diagrammatic reasoning in ZX calculus [26]. At the software architecture level, PyZX abstract the problem using two main data-structures: circuits and graphs.

The circuit class captures the standard representation of quantum protocols as a sequential application of quantum gates. On the other hand, the graph class instantiates the equivalent ZX diagram, which provides a more interesting object for our purpose.

This library has built-in hierarchical simplification strategies. The rules consist of a matcher and a rewriter. At a lower level, the matchers is nothing but a composition of heuristic routines searching for subgraphs satisfying the structural conditions for any of the axiomatic rules of ZX calculus. Once it returns the list of matches, the rewriter applies the corresponding simplifications from equations 6.3, 6.4, 6.5, 6.6, 6.7, and figure 6.1.

This framework will provide the necessary tools for the algorithmic implementation of quantum routines in ZX diagrams, as well as for their topological optimization.

7.1 Logical Verification and Physical Synthesis Duality

PyZX has certain routines, such as `full_reduce`, that simplify the original diagram in a way that could break its physical interpretation in terms of lattice surgery operations. This makes evident the two computational regimes in which ZX diagrams may operate, at least in the context of this work.

On the one hand, ZX calculus represents an intermediate representation of hardware operations where each transformation saves physical resources [46]. For instance, the spiders fusion rule reflects the physical fact that two sequential split (or merge) operations along the same type of boundaries can be unified. This approach helps to reduce the space-time volume of a given protocol.

On the other hand, ZX calculus can be used for logical verification of semantic equivalence. Complex sets of unitary gates can be transformed in a ZX diagram, and its simplification can

manifest semantic equivalence with a more intuitive process. This will be clearly illustrated in the implementation of quantum teleportation.

7.2 Quantum Teleportation

Quantum teleportation is a well-known protocol that transfers an unknown quantum state from one qubit to another [35]. In order to perform this, the protocol consumes two resources: pre-shared quantum entanglement and classical information communication. The standard realisation of quantum teleportation is depicted in figure 7.4.

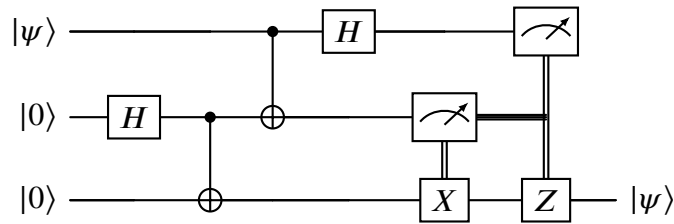


Figure 7.4: Standard quantum teleportation circuit.

Consider Alice has a qubit in an initial quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Alice and Bob share a maximally entangled pair of qubit, for instance

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

so the initial state is:

$$|\psi_0\rangle = |\psi\rangle \otimes |\Phi^+\rangle = \frac{1}{\sqrt{2}}[\alpha(|000\rangle + |001\rangle) + \beta(|100\rangle + |111\rangle)]$$

Alice performs a local interaction with the first two qubits. Firstly, she applies a CNOT, using the unknown qubit as control, and then she applies a Hadamard gate on it. The transmitter's qubits are mapped to a Bell state. The global state is given now by:

$$\begin{aligned} |\psi_1\rangle = & \frac{1}{2}[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) \\ & + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)] \end{aligned}$$

Alice then performs a projective measure on her two qubits, obtaining two classical variables $m_1, m_2 \in \{0, 1\}$. This measure destroys local entanglement, but transports the original unknown state to Bob's qubit. It actually transmits an encrypted version of $|\psi\rangle$, depending on the classical variables:

- If $(m_1, m_2) = (0, 0)$, the state is $\alpha |0\rangle + \beta |1\rangle$ ($|\psi\rangle$).
- If $(m_1, m_2) = (0, 1)$, the state is $\alpha |1\rangle + \beta |0\rangle$ ($X |\psi\rangle$).
- If $(m_1, m_2) = (1, 0)$, the state is $\alpha |0\rangle - \beta |1\rangle$ ($Z |\psi\rangle$).
- If $(m_1, m_2) = (1, 1)$, the state is $\alpha |1\rangle - \beta |0\rangle$ ($ZX |\psi\rangle$).

To recover the original state, we need to transmit the two classical bits to Bob and perform the corresponding Pauli gate. The classical control is useful because it allows us to realize the teleportation protocol without violating connectivity restrictions. However, as PyZX does not have tools to translate classical operations to ZX diagrams, we apply the deferred measurement principle [35] to represent the classical control in terms of quantum gates, as shown in figure 7.5.

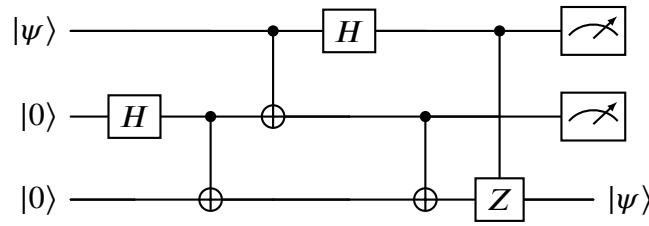


Figure 7.5: Standard quantum teleportation circuit without classical control.

The selection of quantum teleportation as the first routine to implement and optimize does not respond to an arbitrary volition, but rather to its structural relevance due to hardware limitations. The restricted local connectivity of fault-tolerant architectures makes the implementation of logical gates among distant planar patches prohibitive in terms of resources. Therefore, teleportation acquires an undisputed importance, allowing the propagation of quantum information along the processors, as well as being a fundamental ingredient for the transversal implementation of non-Clifford gates [4], surpassing the technical limitations imposed by the Eastin-Knill theorem.

7.3 CHSH Game

The interest of implementing CHSH game (named after Clauser, Horne, Shimony and Holt [9]) resides in two reasons: it provides sufficient proof to demonstrate quantum advantage against any hidden variables theory; and it also justifies the introduction of non-Clifford resources in fault-tolerant architectures.

CHSH game is a problem in information theory that can be modeled as a non-local cooperative game [44]. Two distant players, Alice and Bob, receive each a classical bit of information $x \in \{0, 1\}$ and $y \in \{0, 1\}$. Without any communication, each of them must produce an output bit $a \in \{0, 1\}$ and $b \in \{0, 1\}$, such that they satisfy this boolean relation:

$$a \oplus b = x \cdot y$$

For any strategy restricted to classical phenomena, the probability of winning the game is upper bounded $P_{\text{classic}} \leq 0.75$. We can define a quantum strategy that violates this limit. This requires that Alice and Bob share an entangled pair of qubits, characterized by a Bell state $|\Psi^+\rangle = 1/\sqrt{2}(|00\rangle + |11\rangle)$. Let's discuss the quantum strategy in detail. We start by defining a qubit state vector $|\psi_\theta\rangle$ in terms of an angle θ :

$$|\psi_\theta\rangle = \cos(\theta)|0\rangle + \sin(\theta)|1\rangle$$

The inner product of any two of these vectors is given by:

$$\langle\psi_\alpha|\psi_\beta\rangle = \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta) = \cos(\alpha - \beta)$$

and if we compute the inner product of the tensor product of any of these vectors with the Bell state $|\Phi^+\rangle$:

$$\langle\psi_\alpha \otimes \psi_\beta | \Phi^+\rangle = \frac{\cos(\alpha - \beta)}{\sqrt{2}} \quad (7.2)$$

We also define a unitary matrix U_θ :

$$U_\theta = |0\rangle\langle\psi_\theta| + |1\rangle\langle\psi_{\theta+\pi/2}|$$

Which in a standard convention would be $U_\theta = R_y(-2\theta)$. Alice and Bob share an entangled pair of qubits (A,B) in state $|\Phi^+\rangle$. They receive the physical bit of information, and then perform an operation on their qubits, depending on which classical variable they received. If Alice receives $x = 0$, she will perform U_0 , and $U_{\pi/4}$ for $x = 1$. On the other hand, Bob will perform $U_{\pi/8}$ for $y = 0$ and $U_{-\pi/8}$ for $y = 1$. After they perform this operation, they make a measure of their qubits, and the classical output they obtain are their answers.

For illustrative purposes, let's consider the case $(x, y) = (0, 0)$ using equation 7.2:

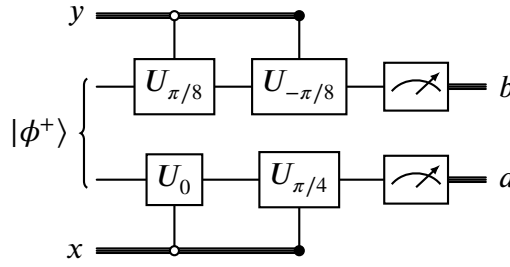
$$\begin{aligned} (U_0 \otimes U_{\pi/8})|\Phi^+\rangle &= |00\rangle\langle\psi_0 \otimes \psi_{\pi/8}| \Phi^+\rangle + |01\rangle\langle\psi_0 \otimes \psi_{5\pi/8}| \Phi^+\rangle \\ &\quad + |10\rangle\langle\psi_{\pi/2} \otimes \psi_{\pi/8}| \Phi^+\rangle + |11\rangle\langle\psi_{\pi/2} \otimes \psi_{5\pi/8}| \Phi^+\rangle \\ &= \frac{\cos\left(-\frac{\pi}{8}\right)|00\rangle + \cos\left(-\frac{5\pi}{8}\right)|01\rangle + \cos\left(\frac{3\pi}{8}\right)|10\rangle + \cos\left(-\frac{\pi}{8}\right)|11\rangle}{\sqrt{2}}. \end{aligned}$$

And the probability of winning the game is:

$$P(a = b) = \cos^2\left(-\frac{\pi}{8}\right) = \frac{2 + \sqrt{2}}{4} \approx 0.85$$

For the remaining three cases, the probability is exactly the same [44], violating the classical limit of $P_{\text{classic}} \leq 0.75$. The realisation of this strategy in standard circuit model can be seen in figure 7.6.

Alice's rotations U_0 and $U_{\pi/4}$ are the identity operation and a Hadamard gate. On the other hand, Bob's rotations $U_{\pi/8}$ and $U_{-\pi/8}$ introduce non-Clifford operations. This enforces

Figure 7.6: CHSH circuit for $(x, y) = (0, 0)$

the use of magic-state distillation, as the Eastin-Knill theorem prevents us from obtaining a transversal realisation of the T gate on a planar surface code. But, as we have already discussed, the injection of magic-states is extremely costly.

When we convert our circuit model to a ZX diagram, PyZX will model the T gate as a phase gadget. This allows us to push the non-Clifford gates through the diagram to find non-local cancelations [27], reducing the T-count and hence the computational cost of the circuit.

7.4 Magic Square Game

The Mermin-Peres magic square game is a pseudo-telepathy scenario that illustrates quantum contextuality [31]. It is once again a cooperative non-local game played by two distant players, Alice and Bob. They are given a 3×3 square which they must fill in a specific way. Alice is assigned a random row, and Bob is given a random column. Each must complete their corresponding cells with values ± 1 , satisfying the following conditions:

- Alice's row must contain an odd number of plus signs.
- Bob's column must contain an odd number of minus signs.
- The cell where Alice's row and Bob's column intersect must contain the same value for both players.

It is easy to see that no classical strategy can produce a victory probability of 100%, as this would imply the construction of a 3×3 matrix where all the rows add up to +1 and all the columns add up to -1, so the global product of the matrix evaluated on the rows would differ from that evaluated on the columns.

However, the use of quantum resources may ensure a deterministically winning strategy. Alice and Bob would need to share two pairs of maximally entangled pairs of qubits. For instance, we could have a global state could described by $|\Phi^+\rangle \otimes |\Phi^+\rangle$. Each of them will take one qubit from each pair.

Instead of building a 3×3 matrix with ± 1 values, we construct a 3×3 grid composed of elements of $\mathcal{P}_2$. We can use this specific arrangement:

$+I \otimes Z$	$+Z \otimes I$	$+Z \otimes Z$
$+X \otimes I$	$+I \otimes X$	$+X \otimes X$
$-X \otimes Z$	$-Z \otimes X$	$+Y \otimes Y$

Now the referee gives them their assignments, and they perform the corresponding measurements on their qubits based on the grid above. As we can see, the three operators in any of the rows commute, so Alice can measure them simultaneously. If you multiply the three operators in any row, they yield $I \otimes I$. The product of Alice's three measurement will always be $+1$. On the other hand, the operators in any of the columns also commute, so Bob can measure simultaneously too. The product of the three operators in any column is $-I \otimes I$, so the product of Bob's measurements will always be -1 .

As the qubits are in a Bell state, when Alice and Bob make the measures corresponding to the same cell in the grid, they will obtain the same result. The three conditions are fulfilled, and hence we have defined a deterministically winning quantum strategy.

The implementation of this game in the unitary circuits paradigm begins by preparing the two pairs of entangled qubits. Then, a classical control decides the specific unitary rotations that Alice and Bob apply on their qubits. To illustrate how these measures are performed, let's consider the case where Alice is assigned the third row and Bob gets the first column. We consider that measurements can be performed in the Z direction, and hence we need to make some necessary rotations to measure the required observables. Alice's first two observables are $-X \otimes Z$ and $-Z \otimes X$. We want to understand the necessary transformation of these operators in the Heisenberg picture in order to obtain the measurable observables $Z \otimes I$ and $I \otimes Z$. Specifically, we want to find a unitary sequence U that maps Alice's two first observables into single-qubit Z measurements:

- $-X \otimes Z \rightarrow Z \otimes I$
- $-Z \otimes X \rightarrow I \otimes Z$

We can clearly see how CZ gate can disentangle the operators:

$$\begin{aligned} CZ(X \otimes Z)CZ &= X \otimes I \\ CZ(Z \otimes X)CZ &= I \otimes X \end{aligned}$$

Now we rotate to the Z -axis:

$$\begin{aligned} (H \otimes H)(-X \otimes I)(H \otimes H) &= -Z \otimes I \\ (H \otimes H)(-I \otimes X)(H \otimes H) &= -I \otimes Z \end{aligned}$$

We finally fix the parity by applying X gates:

$$(X \otimes X)(-Z \otimes I)(X \otimes X) = Z \otimes I$$

$$(X \otimes X)(-I \otimes Z)(X \otimes X) = I \otimes Z$$

Hence, by applying this specific sequence of operators and measuring the first two qubits, we obtain the eigenvalues for the observables $-X \otimes Z$ and $-Z \otimes X$. We also notice that the last observable can be expressed in terms of the first two, $(Y \otimes Y) = (-X \otimes Z) \cdot (-Z \otimes X)$. The product of the three observables of any given row is always $I \otimes I$, and for any column $-I \otimes I$. So Alice's answer for cell 3 can be determined by obtaining the product of the first two eigenvalues, and equivalently for Bob. The specific implementation of this game can be seen in figure 7.7.

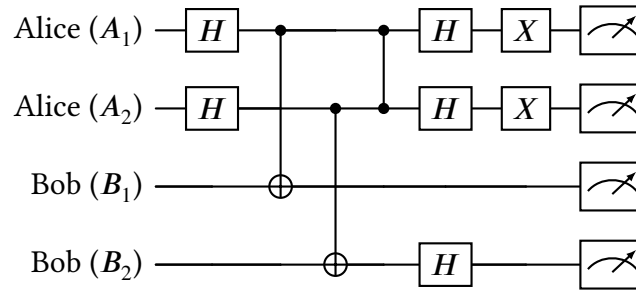


Figure 7.7: Mermin-Peres game circuit implementation for the first column and the third row.

8 | Results

We discuss the PyZX implementations of the three quantum routines presented above. The codes are available in the repository [36].

8.1 Teleportation in PyZX

In order to implement quantum teleportation in PyZX, we need to represent the protocol using the operational description of standard circuit model, as in figure 7.5. Once we instantiate that circuit and translate it to a graph, the resulting diagram will model the realisation of the protocol with lattice surgery operations, considering the necessary classical operations to update the Pauli frame.

Firstly, we instantiate the logical gates of the circuit and convert it to a graph. This conversion does not include neither the preparation of the initial states nor the final measures. For the EPR pair, we need to initialize both logical qubits in the base state $|0\rangle$, which is performed by adding a red degree-1 node at the beginning of each line. Alice's projective measures are performed in an identical way. Finally, the realisation of the classical control admits different options:

- **Partial Quantum Assimilation** We simply instantiate the circuit in figure 7.5. The resulting diagram will have operations among separate qubits, apparently violating the 2DNN connectivity. However, we do not interpret those edges as quantum interactions, but rather as the diagrammatic representation of the classical control after the projective measures. This approach allows the simplification routines to absorb the measures subproducts, while also ensuring the semantic equivalence between the compiled graph and the original circuit.
- **Strict Quantum Assimilation** We could consider both CNOT and CZ gates as genuine quantum interactions. In order to preserve 2DNN connectivity, this approach forces us to include a SWAP gate in the first two qubits. The computational cost of this gate makes this option ignorable, as the interest of the teleportation protocol consists precisely in the possibility of replacing the necessity of SWAP gates.
- **Classical Control Externalization** We suppress the classical control operations in our diagram, faithfully representing the quantum operations involved in the protocol. This option breaks the possibility of performing semantic verification.

If we follow the first approach, we should translate the effects of the projective measures using red degree-1 nodes representing the covectors corresponding with each of the possible results. The deterministic branch where Alice measures both of her qubits in state $|0\rangle$ is shown in figure 8.1. By applying a π phase in the projective nodes, we could represent results in the state $|1\rangle$.

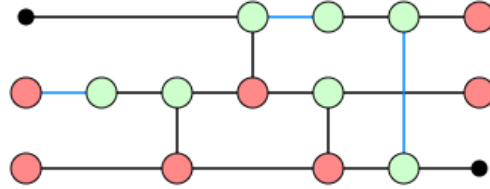


Figure 8.1: ZX diagram corresponding to a standard quantum teleportation when Alice measures $(0, 0)$.

For each of the diagrams $D_{ZX}(m_0, m_1)$, $m_0, m_1 \in \{0, 1\}$, we perform the simplification routine of `full_reduce`. The result is that the four diagrams collapse to the identity channel I , as shown in figure 8.2. This constitutes the validation of the implemented protocol.

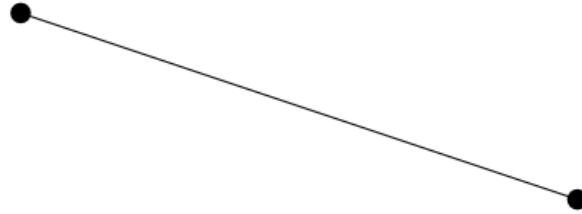


Figure 8.2: Identity channel resulting from the simplification of the teleportation diagrams

The diagram in figure 8.1, ignoring the interactions modeling the feedforward, can be viewed in terms of lattice surgery operations (given that the Pauli frame is correctly updated). We can break the simplification routine in each of its steps, and look for simplifications of the protocol that do not break the physical interpretation of the diagram as composition of primitives of lattice surgery. The four diagrams yield the same result prior to the total collapse to the identity channel, as shown in figure 8.3.

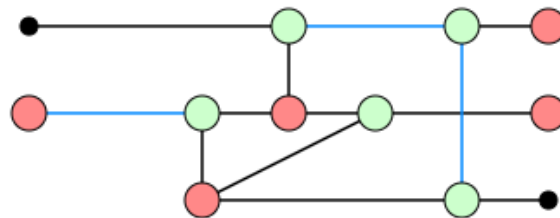


Figure 8.3: Simplification of the teleportation diagram when Alice measures $(0, 0)$.

The diagram has a reduced number of degree-3 nodes, but we notice that the simplification comes from the topological assimilation of part of the classical control, specifically the

controlled X operation, contracting it with the previous quantum CNOT operation. Therefore, the space-time volume of the protocol is not reduced.

In order to compile the protocol we try by externalizing the classical control, exclusively considering the quantum operations. The corresponding diagram is shown in figure 8.4

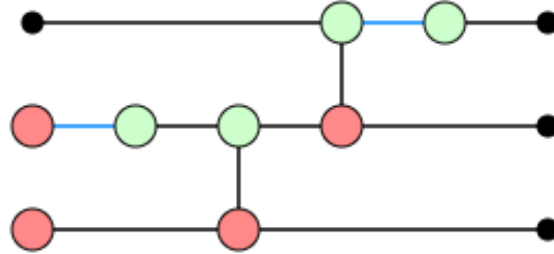


Figure 8.4: Teleportation protocol without the classical control operations.

Once again, instead of using the `full_reduce` routine, we break it into each of its simplification steps. In this case, the complete routine yields an interesting diagram:

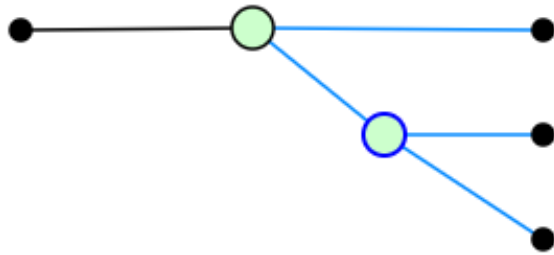


Figure 8.5: Simplified teleportation protocol without the classical control operations.

Previously, the classical control operations allowed the simplification routine to collapse the diagram to the logical identity. Now, the result of the routine can be put in a direct isomorphism with lattice surgery operations. We have a unique entry node representing the original state $|\psi\rangle_L$ encoded in a logical qubit constituted by three logical patches, and through two smooth splits we obtain three highly entangled logical qubits. By performing the projective measures on the two first qubits, and the corresponding Pauli operations on the third, we obtain the original state $|\psi\rangle_L$ in the last qubit.

If we compare figures 8.4 and 8.5, we see that the maximum layer index is reduced from 4 to 3. This represents a huge improvement in the execution time of the algorithm. Also, the pathwidth is reduced from 4 to 3, optimizing the number of logical patches necessary for the implementation.

The physical interpretation presents a problem: the logical patch where we want to teleport the unknown state is originally merged in the logical qubit where that unknown state is encoded. If we want to separate the original state and the target qubit, we can find another implementation in a previous step of the simplification routine, represented in figure 8.6

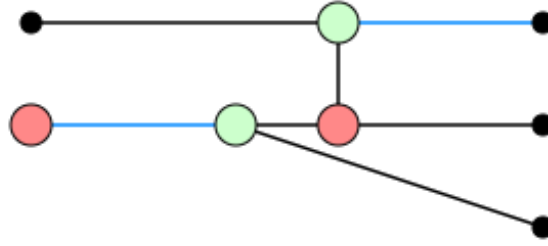


Figure 8.6

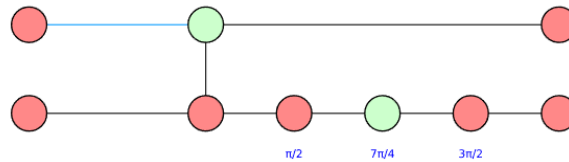
Now, the original state represents an independent entry wire, while the two entangled logical qubits are firstly merged and prepared in the $|0\rangle_L$ state and then separated through a smooth split. The first two qubits are then entangled. This realisation of the algorithm reduces the maximum layer index from 4 to 3, and the pathwidth is not reduced. It represents a smaller optimization, but a more useful realisation.

8.2 CHSH Game in PyZX

In contrast to the evaluation of the teleportation protocol—where the compiler’s objective resided in preserving open logical wires to demonstrate the invariance of the identity channel $\mathcal{I}$ —the topological instantiation of the CHSH game is formulated as the strict evaluation of a closed tensor network.

In the diagrammatic formulation using PyZX, the classical variables of the referee ($x, y \in \{0, 1\}$) modulate the local rotation sequences, while the joint measurement outcomes ($m_0, m_1 \in \{0, 1\}$) are imposed as strict boundary conditions on the output ports. These are instantiated via arity-1 tensors of type X (red spiders) with phases $m_i\pi$. This total closure of the topological boundaries, as shown in figure 8.7, transforms the diagram from a linear matrix operator $\mathcal{H} \rightarrow \mathcal{H}$ into a rank-0 tensor, i.e., a complex scalar. Semantically, the closed graph represents the geometric isomorphism of the exact probability amplitude for each branch:

$$\mathcal{A}_{x,y}(m_0, m_1) = \langle m_0 m_1 | U_{x,y} | 00 \rangle \quad (8.1)$$

Figure 8.7: ZX diagram corresponding to $(x, y) = (0, 0)$ and $(m_0, m_1) = (0, 0)$

By iterating programmatically over the 16 possible permutations dictated by the combinations of (x, y) and (m_0, m_1) , and subjecting each resulting diagram to the algebraic simplification heuristics, a singular analytical phenomenon is observed: the tensor network collapses entirely into a visually empty diagram.

Far from representing an algorithmic error or a loss of information, the topological annihilation of all nodes and edges constitutes the geometric manifestation of a successful **strong simulation**. The rewriting engine of the Frobenius subalgebra contracts the entirety of the lattice surgery isometries alongside the non-Clifford rotations ($\pm\pi/4$), fully resolving the underlying system of algebraic equations. The empty diagram signifies that the ZX-calculus has successfully evaluated the full trace of the tensor network, encapsulating the complex amplitude $\mathcal{A}_{x,y}(m_0, m_1)$ within the global scalar factor of the remaining graph.

This absolute collapse validates the compiler’s capability to act as a deterministic evaluator of multipartite systems. By mathematically recovering the scalar hidden within each “empty diagram”, one obtains direct access to the exact marginal probability distribution $P(m_0, m_1 | x, y) = |\mathcal{A}_{x,y}(m_0, m_1)|^2$ [28] [41]. This extraction is a fundamental prerequisite for the subsequent certification of the analytical violation of the local hidden-variable (LHV) bounds within the surface code topology.

Indeed, once we extract the marginal probability amplitude from each of the diagrams and compute the probability of victory for each pair of classical variables (x, y) , we obtain $P_{\text{quantum}} \approx 0.85$, violating the LHV bounds.

Now we can connect a key element of quantum information theory with the hardware realities of fault-tolerant architectures by discussing the implementation and compilation of the protocol. As we did in the teleportation diagram, we exclude the measurement projections:

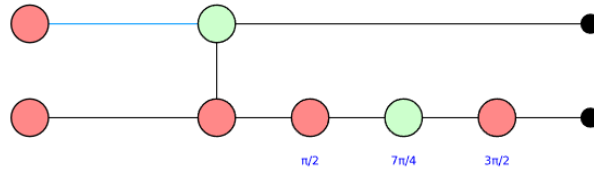


Figure 8.8: ZX diagram for the implementation of CHSH game, corresponding to $(x, y) = (0, 0)$.

Bob’s measurements require rotations in the Y axis with angles of $\pm\pi/4$, which are not elements of the Clifford group. The simplification routines cannot annihilate them, and this is not an avoidable hardware limitation but rather a consequence of the Eastin-Knill theorem, as our fault-tolerant architecture necessarily lacks a universal set of transversal quantum gates.

Consequently, the realisation of the CHSH protocol forces the compiler to invoke magic state injection protocols [37], which are extremely costly, as we have already discussed. The greatest contribution to the space-time volume for this realisation comes from the requirement of non-Clifford resources, not from generating the initial entanglement.

Even so, we perform our simplification routine to each of the four possible diagrams. The procedure yields a significant reduction of Clifford gates, depicted in figure 8.9, which, as we have already stated, does not imply a huge reduction of execution time, but it is still an optimization. The topological depth for each of the four branches is reduced from 5 to 3. The pathwidth remains the same. The translation of the diagram to lattice surgery primitives is immediate:

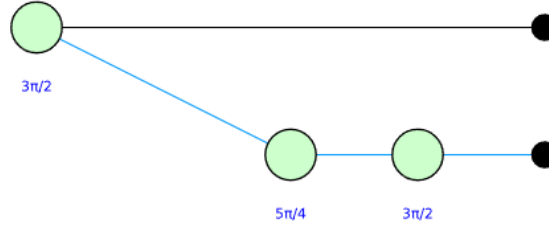


Figure 8.9: Simplification of the ZX diagram for the implementation of CHSH game, corresponding to $(x, y) = (0, 0)$.

- We prepare a logical qubit in the state $|0\rangle + e^{i3\pi/2} |1\rangle$.
- A smooth split is performed, and then we apply a T gate on Bob's qubit.
- A final S gate is performed on Bob's qubit, and after that we measure each qubit.

The realisation in figure 8.9 could not be feasible due to the fact that we are preparing an initial state that does not belong to the computational basis. Another realisation with a better physical translation is shown in 8.10

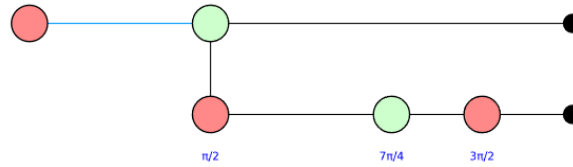


Figure 8.10: Simplification of the ZX diagram for the implementation of CHSH game, corresponding to $(x, y) = (0, 0)$, and with preparation of initial state in the computational basis.

The topological depth is reduced from 5 in figure 8.8 to 4 in figure 8.10.

8.3 Magic Square Game in PyZX

The evaluation of the Mermin-Peres magic square game introduces the verification of quantum contextuality and pseudo-telepathy in fault-tolerant architectures. Unlike the CHSH protocol, which pursues a statistical limit, the Mermin-Peres game demands the verification of a strict deterministic victory, overcoming the classical bound of $8/9$ for this problem.

The protocol requires Alice and Bob to share two pairs of entangled qubits. The instantiation of the circuit begins by creating those two pairs. Next, each player performs the operations corresponding to their observables for the determined row and column. This produces nine different diagrams, with three different subdiagrams for each player's qubit.

The validation of the strategy requires a similar approach to the CHSH game. We firstly notice that, once we set boundary conditions in the output wires, placing red degree-1 nodes corresponding to measurement projections, as shown in 8.11, the simplification routine `full_reduce` annihilates the tensor network, collapsing towards an empty graph.

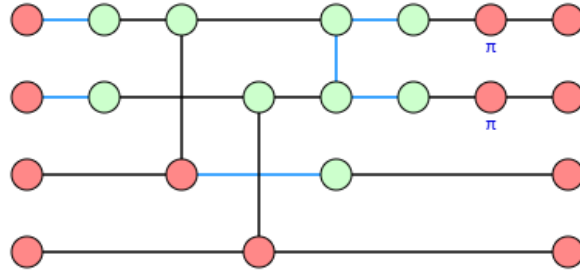


Figure 8.11: ZX diagram for row 3 and column 0, representing the circuit in 7.7, with boundary conditions corresponding to the covector $\langle 0000|$.

As we saw in the CHSH game, this is an expected consequence from the fact that the graph does not have open wires, meaning it represents a rank-0 tensor (a scalar). The collapse allows us to evaluate the protocol, as the probability amplitude of each stochastic branch is encapsulated within the global scalar factor of the diagram.

Hence, we extract those amplitudes and obtain the output probabilities for each selection of row and column. By comparing those outputs with the winning conditions, we are able to compute the success probability of the strategy. The application of this method yields us a probability of victory of 1.0, verifying the deterministically winning strategy.

The translation of the Mermin-Peres game, from its abstraction in quantum information theory towards its realisation in a fault-tolerant architecture, requires some modifications to the previous diagrams, attending to the spatial limitations imposed by the hardware.

The initial preparation of the Bell pairs assume global connectivity, which is manifested in the first two CNOT gates of figure ???. The physical execution of those gates is not viable under the hardware restrictions we are considering for our architecture. In order to fix this and preserve 2DNN connectivity, we are forced to use a SWAP gate, which constitutes an important operative bottleneck. The realisation of the protocol will then begin by the preparation of two Bell pairs, and then we will perform a SWAP gate to interchange one qubit from each pair.

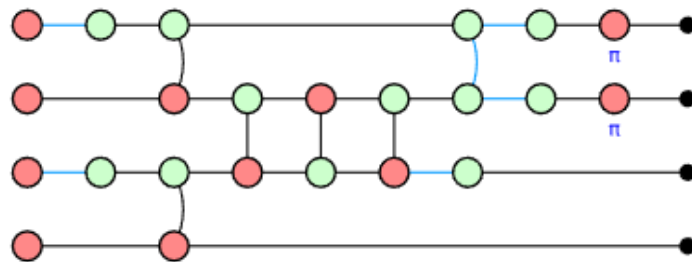


Figure 8.12: ZX diagram for row 3 and column 0 with 2DNN connectivity.

Compared to the implementation of the CHSH game, the Mermin-Peres games does not require the use of non-Clifford gates, thus suppressing the necessity of magic-states distillation. The diagram has a higher number of nodes and edges, so the obtention of its topological depth requires the algorithmic approach for the orientation of vertical edges presented in chapter 7. This case, as well as the other eight diagrams, constitute trivial cases for the execution

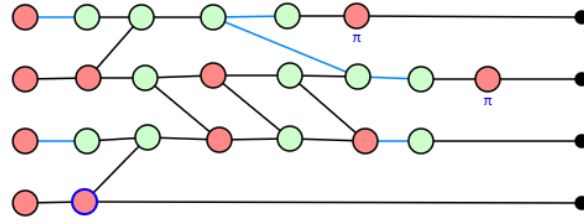


Figure 8.13: Visual representation of the orientation of edges for row 3 and column 0. Time flows from left to right.

of the algorithm, as there are no vertical edges connecting nodes with the same depth. We can build some intuition of the method by looking at figure 8.13. For the trivial cases where $L_{in}(u) = L_{in}(v)$, we just need to draw the vertex in an explicit layer distribution, so each of the vertical edges will be drawn as a diagonal and will be oriented in the direction of time flux.

The next step is to perform the simplification routine on each diagram. The simple execution of `full_reduce` produces graphs without possible translation to lattice surgery operations, so we once again use the stepped simplification routine. We then select the last intermediate optimization that admits an interpretation as a physical realisation of the protocol. The result for the diagram in figure 8.12 can be seen in figure 8.14.

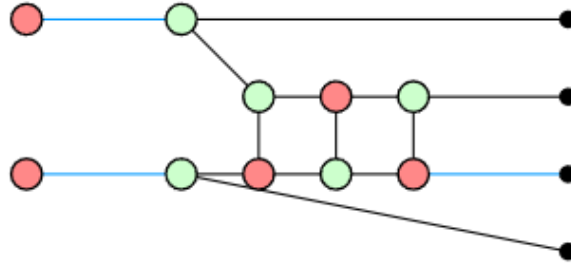


Figure 8.14: Simplified graph for row 3 and column 0.

We can compare the computational cost metrics for the primitive and the optimized diagrams. The data in table 8.1 shows that the simplification method reduces the depth of our diagrams in average around an 18%, and the pathwidth is reduced on average approximately in a 9%. Out of the 9 diagrams, 7 of them had a reduced depth once they were simplified, and 5 of them reduced their pathwidth. All the graphs improved at least one of the metrics after the optimization.

Table 8.1

Metric	(0,0)	(0,1)	(0,2)	(1,0)	(1,1)	(1,2)	(2,0)	(2,1)	(2,2)
Original Depth	7	6	8	7	6	8	8	8	8
Original Pathwidth	6	6	6	6	6	6	6	6	7
Optimized Depth	5	6	6	6	6	6	6	6	7
Optimized Pathwidth	5	5	6	5	5	6	6	6	6

9 | Conclusions

Throughout this dissertation, we have addressed one of the fundamental challenges for the viability of quantum computations: the fragility of quantum information against decoherence and environmental noise. For the mitigation of the negative effects of such phenomena, fault-tolerant quantum computing relies on the theory of Quantum Error Correction, and, specifically, on topological codes.

The treatment of said structures for the implementation of transversal gates without violating connectivity restrictions require the introduction of lattice surgery, a set of fundamental procedures that allow us to construct an operational model for fault-tolerant quantum computation.

The execution of the primitive operations of lattice surgery implies projective measurements on a given set of physical qubits, and the results of said operations alter the dimensionality of the code subspace. Therefore, lattice surgery breaks the unitary evolution of quantum system, hence making the standard circuit model for quantum computation unable to faithfully represent the dynamics of the logical qubits under the actions of merges and splits.

We have seen how ZX calculus can perform the role of graphical language for fault-tolerant computation, constituting the analytical core of this work. ZX calculus does not only provide the visual support to compile quantum protocols into lattice surgery operations, but also the algebraic tools for the validation and optimization of said protocols.

The validation of the logical architectures has been materialized through the PyZX ecosystem. For the teleportation protocol, the set of diagrams parametrized by the measurement outcomes has been evaluated. The graph theoretic reduction routine has collapsed this set of diagrams towards an identity channel, thus verifying the functionality of quantum teleportation in a fault-tolerant architecture, which constitutes a transcendental protocol for the transversal execution of non-Clifford gates, as well as for the execution of operations requiring the interaction of distant qubits.

Analogously, the evaluation of the CHSH game has validated the entanglement and quantum correlations. The magic square game has demonstrated the possibility of contextuality in this paradigm.

Finally, the execution of optimization routines has yielded a significant reduction of the space-time volume for the different protocols we have discussed, materializing the optimization potential of ZX calculus.

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A | Algebraic Foundations of ZX Calculus

ZX calculus has been introduced primarily as a syntactic and operational tool. In equation 6.1, the Z and X spiders are defined as projectors over the computational and diagonal bases, respectively. On the other hand, the topological meta-rule that allows diagrammatic deformations as well as the spider fusion rule from equation 6.3 might appear as convenient but some-what ad-hoc graphical heuristics for tensor contraction. To grasp the analytical power of ZX calculus, we should transcend this merely syntactic view, by presenting a formal mathematical treatment, matching the rigour of chapters 4 and 5.

Categorical Foundations

| Definition A.1. *A monoidal category is a category C equipped with a tensor product $\otimes$ that combines object A and B into composite objects $A \otimes B$ and morphisms f and g into parallel operations $f \otimes g$, with a monoidal unit $I \in \text{Obj}(C)$ [13].*

Monoidal categories are interesting from a computational point of view as they allow us to characterize the parallel evolution of different systems. ZX diagrams can be formally defined as morphisms within a specific monoidal category denoted as **ZX**. The objects of said category are the natural numbers $n, m \in \mathbb{N}$, representing the number of input and output wires of a diagram. The tensor product is defined as an integer addition $n \otimes m = n + m$ [45]. The ZX diagrams are the morphisms of this category, meaning that a diagram with n inputs and m outputs is nothing but a morphism $D : n \rightarrow m$

ZX calculus treats inputs and outputs interchangeably, which is extremely useful but also seemingly unjustified. This is actually obtained by bending wires, but to understand it we need to introduce additional structure. We define two special morphisms called cups and caps and represented as [25]:

$$\left(\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right) = |00\rangle + |11\rangle \quad \left(\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right) = \langle 00| + \langle 11| \quad (\text{A.1})$$

Intuitively, cups and caps bend an input wire to become an output, and vice versa. Cups satisfy the following relation with spiders:

$$n \left\{ \begin{array}{c} \vdots \\ \vdots \end{array} \right\} \alpha \left\{ \begin{array}{c} \vdots \\ \vdots \end{array} \right\} m = n \left\{ \begin{array}{c} \vdots \\ \vdots \end{array} \right\} \alpha \left\{ \begin{array}{c} \vdots \\ \vdots \end{array} \right\} m \quad (\text{A.2})$$

Which can be easily verified in bra-ket notation:

$$\left(\langle 00| + \langle 11| \right) \left(\underbrace{|0 \dots 0\rangle}_{m+1} \underbrace{\langle 0 \dots 0|}_n + e^{i\alpha} \underbrace{|1 \dots 1\rangle}_{m+1} \underbrace{\langle 1 \dots 1|}_n \right) = \underbrace{|0 \dots 0\rangle}_m \underbrace{\langle 0 \dots 0|}_{n+1} + e^{i\alpha} \underbrace{|1 \dots 1\rangle}_m \underbrace{\langle 1 \dots 1|}_{n+1}$$

For the symmetry relation of caps and spiders, we have:

$$n \left\{ \begin{array}{c} \vdots \\ \vdots \end{array} \right\} \alpha \left\{ \begin{array}{c} \vdots \\ \vdots \end{array} \right\} m = n \left\{ \begin{array}{c} \vdots \\ \vdots \end{array} \right\} \alpha \left\{ \begin{array}{c} \vdots \\ \vdots \end{array} \right\} m \quad (\text{A.3})$$

Which once again is immediate if we use bra-ket notation:

$$\left(\underbrace{|0 \dots 0\rangle}_m \underbrace{\langle 0 \dots 0|}_{n+1} + e^{i\alpha} \underbrace{|1 \dots 1\rangle}_m \underbrace{\langle 1 \dots 1|}_{n+1} \right) \left(|00\rangle + |11\rangle \right) = \underbrace{|0 \dots 0\rangle}_{m+1} \underbrace{\langle 0 \dots 0|}_n + e^{i\alpha} \underbrace{|1 \dots 1\rangle}_{m+1} \underbrace{\langle 1 \dots 1|}_n$$

Equations A.2 and A.3 also hold for Z spiders and formally prove that bending wires is perfectly valid. A symmetric monoidal category with cap and cup-like structures is a compact closed category, which is the principal structure of categorical quantum information [2]. It is this characterisation what proves the fact that we can move the generators around the diagram without altering its semantics [24].

Definition A.2. A dagger category is a category C equipped with an involutive, identity-on-objects, contravariant functor $\dagger : C \rightarrow C$. If C is a symmetric monoidal category and $\dagger$ preserves the symmetric monoidal structure, then C is a dagger symmetric monoidal category [38].

We indeed have such a functor in $\mathbf{ZX}$, as it consists of flipping any ZX diagram horizontally and changing the signs of all the phases. Therefore, $\mathbf{ZX}$ is a dagger compact closed category. The deformation and inversion of the diagrams that this category allows is the foundation for the meta-rule of ZX calculus: only connectivity matters.

The semantic meaning of ZX diagrams reside in the category of finite-dimensional complex vector spaces, specifically the subcategory **Qubit**[45]. The objects of **Qubit** are also the natural numbers, and we interpret $n \in \mathbb{N}$ as the vector space $\mathbb{C}^{2^n}$. The tensor product is the regular

tensor product of vector spaces. The morphisms in this category are linear maps $f : \mathbb{C}^{2^n} \rightarrow \mathbb{C}^m$. We define the interpretation of a ZX diagram as a functor $[[\cdot]] : \mathbf{ZX} \rightarrow \mathbf{Qubit}$. For the objects, we simply have $[[n]] = n$, and for a diagram D we set $[[D]]$ to be the linear map that the diagram represents.

Theorem 6.1 proves that the interpretation of ZX diagrams is a full functor, i.e., for any $f : \mathbb{C}^{2^n} \rightarrow \mathbb{C}^m$ there is a diagram $D : n \rightarrow m$ such that $[[D]] = f$. However, the interpretation functor is not faithful, meaning that the equality $[[D]] = [[D']]$ does not imply $D = D'$. This is fairly obvious, as the rewrite strategies of chapter 5 yield different diagrams representing the same linear map. We can define the quotient category $\mathbf{ZX}/\sim$, where $\sim$ is an equivalence relation such that two diagrams that can be rewritten into one another are identified. If the rewrite ruleset we use to define the equivalence relation is complete (meaning that any two diagrams representing the same linear map can be rewritten into one another), then the interpretation functor of the quotient category is indeed faithful [45]. This may seem trivial, but completeness is a central topic in ZX calculus and the discussion proves that it can be formally viewed as a property of the functor $[[\cdot]]$.

Monoids, comonoids and Frobenius Algebra

Definition A.3. In a monoidal category $(C, \otimes, I)$, a monoid is an object $A \in \text{Obj}(C)$ with two structural morphisms: a multiplication operation $\mu : A \otimes A \rightarrow A$ and a unit morphism $\eta : I \rightarrow A$ such that they satisfy the axiom of associativity $\mu \circ (\mu \otimes \text{id}_A) = \mu \circ (\text{id}_A \otimes \mu)$ and the left and right unit laws [40].

In quantum information, monoids model processes in which two systems are merged into a unified subspace. By inverting the directions of the morphisms, one could define a dual concept, known as a comonoid:

Definition A.4. In a monoidal category $(C, \otimes, I)$, a comonoid is an object $A \in \text{Obj}(C)$ with two structural morphisms: a comultiplication operation $\delta : A \rightarrow A \otimes A$ and a counit $\epsilon : A \rightarrow I$ such that they satisfy the axiom of coassociativity $(\delta \otimes \text{id}_A) \circ \delta = (\text{id}_A \otimes \delta) \circ \delta$ and the corresponding counit laws.

If a given object has a monoid and a comonoid, we could compose their corresponding morphisms and study their interaction. We would like to find that this interaction respects some symmetry, specifically, that the sequential composition of a comultiplication and a multiplication should be topologically equivalent to the same composition applied to parallel asymmetric channels. This is captured in the Frobenius condition, that defines what a Frobenius algebra is:

Definition A.5. In a monoidal category $(C, \otimes, I)$, a Frobenius algebra is an object $A \in \text{Obj}(C)$ equipped with a monoid (A, μ, η) and a comonoid (A, δ, ϵ) , such that the Frobenius condition holds:

$$(\text{id}_A \otimes \mu) \circ (\delta \otimes \text{id}_A) = \delta \circ \mu = (\mu \otimes \text{id}_A) \circ (\text{id}_A \otimes \delta) \quad (\text{A.4})$$

Intuitively, a graph consisting of multiplications and comultiplications can be topologically deformed without altering the underlying operator. We could demand some additional axiomatic restrictions [11]:

1. Commutativity: The monoid must be commutative, $\mu \circ \sigma_{A,A} = \mu$, where $\sigma_{A,A}$ is the symmetry isomorphism of the category.
2. Special Condition: The composition of comultiplication and multiplication is the identity $\mu \circ \delta = \text{id}_A$.
3. Dagger structure: The structure of the comonoid should be obtained by the application of the dagger functor on the monoid, $\delta = \mu^\dagger$ and $\epsilon = \eta^\dagger$.

If a Frobenius algebra satisfies this three conditions, it will be called a dagger-special commutative Frobenius algebra, or $\dagger$ -SCFA. It is easy to note that the Z and X generators of the ZX calculus are exactly $\dagger$ -SCFAs. The spider fusion rule from equation 6.3 naturally emanates from the axioms of Frobenius algebras, ensuring the associativity and coassociativity, as well as the Frobenius condition.

The $\dagger$ -SCFA characterises a single spider, Z or X . The interaction between both spiders is called a bialgebra and it consists of a monoid and a comonoid corresponding to each algebra and satisfying a commutation relation that is equivalent to equation 6.7 [45]. If we have both algebras, and the sub-index of each morphism denotes the generator it is associated with, $\delta_Z \circ \mu_X = \mu_X \circ \delta_Z$ and vice versa.

If we define an antipode morphism $S : A \rightarrow A$ satisfying the Hopf law:

$$\mu \circ (\text{id} \otimes S) \circ \delta = \mu \circ (S \otimes \text{id}) \circ \delta = \eta \circ \epsilon \quad (\text{A.5})$$

where μ denotes the multiplication and δ the comultiplication of a bialgebra, we elevate it to a Hopf algebra [13]. Two $\dagger$ -SCFAs that are Hopf algebras in both possible combinations of monoid and comonoid is a Hopf-Frobenius algebra. Z generators and X generators constitute a Hopf-Frobenius algebra, where the antipode is the identity.

A fundamental result in categorical quantum mechanics states a bijective correspondence between orthonormal basis in the category of finite dimensional Hilbert spaces **FdHilb** and $\dagger$ -SCFAs [12]. This relation alongside the Hopf law constitutes the regular complementarity of mutually unbiased bases, i.e., given two orthonormal bases $Z = \{|z_i\rangle\}$ and $X = \{|x_j\rangle\}$ that are mutually unbiased ($|\langle z_i | x_j \rangle|^2 = 1/d \forall i, j$ where d is the dimension of the Hilbert space), maximal information about the outcome of a Z measurement yields minimal information about the outcome of a X measurement [25]. What we have discussed can be seen as the formal generalization of a well-known result of quantum mechanics in the language of monoidal categories theory, providing a physically coherent foundation for ZX calculus.

There is an obvious correspondence between lattice surgery operations and the structural morphisms of $\dagger$ -SCFAs. Lattice splits act as a monoidal comultiplication on **FdHilb**, $\delta : V \rightarrow V \otimes V$, and merges as a multiplication $\mu : V \otimes V \rightarrow V$. This perfectly matches with the correspondence between lattice surgery and ZX calculus discussed in chapter 5.